



SGITM OnyxTM 3000 Series Systems

Quick Reference

SGI Confidential & Proprietary Information - For Internal Recipients Only

This document contains proprietary and confidential information of Silicon Graphics, Inc., intended for internal recipients only. The contents of this document may not be disclosed to third parties, copied, or duplicated in any form, in whole or in part, without the prior written permission of Silicon Graphics, Inc.

Limited Rights Legend

The electronic (software) version of this document was developed at private expense; if acquired under an agreement with the USA government or any contractor thereto, it is acquired as "commercial computer software" subject to the provisions of its applicable license agreement, as specified in (a) 48 CFR 12.212 of the FAR; or, if acquired for Department of Defense units, (b) 48 CFR 227-7202 of the DoD FAR Supplement; or sections succeeding thereto. Contractor/manufacturer is Silicon Graphics, Inc., 1600 Amphitheatre Pkwy. 2E, Mountain View, CA 94043-1351.

Silicon Graphics, the Silicon Graphics logo, InfiniteReality, IRIX, Onyx, Onyx2, and OpenGL are registered trademarks and SGI, InfiniteReality2, InfiniteReality3, MultiLink, NUMALink, Origin, Stereo View and SuperWide are trademarks of Silicon Graphics, Inc.

Genelec is a trademark of Genelec OY. Panasonic is a trademark of Matsushita Electric Corporation of America. Sony is a trademark of Sony Electronics, Inc. Spirit Datacine and Philips are trademarks of Koninklijke Philips Electronics N.V. X Window System is a trademark of The Open Group.

**SGI Onyx 3000 Series Systems Quick Reference
Document Number 108-0304-001**

**SGI
Mountain View, California**

Contents

About this Booklet	ix
Notational Conventions	ix
Supporting Documentation	x
 1. Onyx 3000 and G-brick Components	1-1
1.1 New Hardware Terminology	1-2
1.2 Standard Onyx 3000 Series Components	1-2
1.3 Onyx 3000 Series Component Block Diagrams	1-3
1.4 Onyx 3000 Series Rack Drawings	1-5
1.5 The G Brick	1-7
1.5.1 G Brick vs. KCAR (Onyx2)	1-7
1.5.2 G-brick Dimensions	1-7
1.5.3 G-brick Midplane	1-8
1.5.4 G-brick Components	1-8
1.5.5 G-brick-compatible Graphics Board Sets	1-9
1.6 DG5 Boards	1-10
1.6.1 DG5 Connector Overview	1-11
1.6.2 DG5-8 Notes	1-12
1.6.3 DG5-DPLEX Notes	1-12
1.6.4 DG5-DDO2 Notes	1-12
1.6.5 DG5-GVO Notes	1-12
1.6.6 DG5-TVO Notes	1-13
1.6.7 DG5 Board Drawings	1-13
1.6.8 DG5 Monitor Pinouts	1-16
1.6.9 Calculating DG5 Output Bandwidth	1-17
1.7 Ktown2 Board	1-17
1.8 RM Boards	1-18
1.9 GE16-4 Board	1-19
1.10 Onyx 3000 Series Graphics Displays	1-20
1.10.1 Monitor Cables	1-20
1.10.2 Onyx 3000 Series System with 1600SW Flat Panel Display	1-22
1.11 Keyboard, Mouse, and Hubs	1-23
1.11.1 The Standard Hub	1-23
1.11.2 The Extender Hubs	1-24
1.11.3 Mouse and Keyboard Controller Numbers (Device IDs)	1-26
1.11.4 Configuring Input Devices	1-27
1.12 Onyx 3000 Series Audio Components	1-27
1.12.1 Audio PCI Card	1-27
1.12.2 Speakers	1-29
1.13 SIO PCI Card	1-31
1.14 Digital Media XIO Options	1-31
1.14.1 DIVO Notes	1-31
1.14.2 DIVO-DVC Notes	1-32

1.14.3	XT-HD Notes	1-32
1.15	L1 Controller	1-33
1.15.1	L1 Hardware	1-33
1.15.2	Basic L1 Controller Functions	1-34
1.16	Part Numbers	1-35
2.	Onyx 3000 Series Configuration.....	2-1
2.1	Configuration Overview	2-2
2.2	Basic Configuration Rules	2-2
2.3	Example Rackmount Graphics Configurations	2-3
2.4	Graphics Rack Layout and Bay Numbering.....	2-4
2.5	Graphics Rack Placement	2-5
2.6	Assigning Pipe Numbers	2-6
2.7	Suggested Pipe Configuration	2-7
2.8	Checking Actual Pipe Numbering	2-9
2.9	Pipe and X Server Numbering.....	2-9
2.9.1	hostname:0.0 Described	2-9
2.9.2	The Xservers File: Configuring Graphics Seats and Assigning Keyboard and Mouse Control	2-10
2.9.3	Pipe Failure and Pipe Numbering	2-11
2.10	Recommended I/O-bricks-to-G-brick Connections.....	2-12
2.11	Upgrade Notes	2-13
2.12	Power Cabling.....	2-14
2.13	Keyboard and Mouse Configuration.....	2-14
3.	Commands, Tools, and Troubleshooting.....	3-1
3.1	Starting and Stopping Graphics	3-2
3.2	Graphics Maintenance Tools	3-2
3.2.1	Diagnostic Testing (<i>irsaudit</i>)	3-2
3.2.2	Video Format Combiner (<i>ircombine</i>).....	3-3
3.2.3	Maintaining the EEPROM (<i>ireeprom</i>)	3-4
3.2.3.1	Restoring or Upgrading the EEPROM Code	3-4
3.2.3.2	Restoring a Default Combination.....	3-5
3.2.4	Changing the Video Format with <i>setmon</i>	3-5
3.3	Displaying the Graphics Subsystem Information.....	3-6
3.4	Displaying the Graphics Topology	3-6
3.5	Running Boundary Scan on Graphics Hardware	3-6
3.6	L1, L2, PROM Monitor, and POD commands.....	3-6
3.7	Troubleshooting Graphics.....	3-7
	Notes.....	11
	Index	1

Figures

Figure 1-1	Single-rack Graphics Block Diagram Example.....	1-3
Figure 1-2	2-pipe Onyx 3000 Series System Block Diagram.....	1-4
Figure 1-3	Single-rack System Components (Front View).....	1-5
Figure 1-4	Major Components of the Single-rack System (Rear View)	1-6
Figure 1-5	G-brick (and Onyx2 Rackmount) Midplane Rear View.....	1-8
Figure 1-6	Graphics Pipe Board Locations.....	1-9
Figure 1-7	DG5-2 Graphics Board with No Options	1-13
Figure 1-8	DG5-8 Board	1-14
Figure 1-9	DG5-2 with Optional DPLEX Daughter Card	1-14
Figure 1-10	DG5-2 with Optional DDO2 Daughter Card	1-15
Figure 1-11	DG5-2 with Optional GVO Connectors.....	1-15
Figure 1-12	DG5-2 with Optional TVO Daughter Card	1-16
Figure 1-13	DG5 13W3 Monitor Pinouts	1-16
Figure 1-14	Ktown2 Board	1-17
Figure 1-15	RM7 Board	1-18
Figure 1-16	GE16-4 Board	1-19
Figure 1-17	13W3 Monitor Cable and Adapters.....	1-21
Figure 1-18	Primary I-brick Keyboard/Mouse/Hub Connectors	1-23
Figure 1-19	Standard Hub	1-24
Figure 1-20	LEX and REX Hubs.....	1-24
Figure 1-21	Standard Configuration for Extender, Hub, and Dual Seats.....	1-25
Figure 1-22	Example Device IDs and Paths in /etc/ioconfig.conf.....	1-26
Figure 1-23	Audio PCI Card and Breakout Cables.....	1-28
Figure 1-24	Speaker Cabling	1-29
Figure 1-25	Speaker Front.....	1-30
Figure 1-26	Rear Controls and Connectors (Right Speaker).....	1-30
Figure 1-27	G-brick L1 Display, LEDs, and Power Button	1-33
Figure 1-28	Location of L1 Controller Logic.....	1-33
Figure 1-29	G-brick L1 to System L2 Cabling Example	1-34
Figure 2-1	Single-rack, Single-pipe Configuration Example.....	2-3
Figure 2-2	Three-rack, Three-pipe (4RM) Configuration Example	2-3
Figure 2-3	Bay (Slot) Numbering and Rack Layout.....	2-4
Figure 2-4	Onyx 3000 Series Rack Placement	2-5
Figure 2-5	Example Pipe Sequencing.....	2-6
Figure 2-6	Suggested Pipe Sequence with All Pipes in Adjacent Racks.....	2-7
Figure 2-7	Suggested Pipe Sequence with Graphics Racks on Both Ends ...	2-8
Figure 2-8	Numbering Before and After Pipe Failure at Boot – Example .	2-11
Figure 2-9	I/O Brick and Pipe Relation.....	2-12
Figure 2-10	Power Cabling.....	2-14

Tables

Table 1-1	Comparison: G Brick and KCAR.....	1-7
Table 1-2	G-brick Technical Specifications	1-7
Table 1-3	InfiniteReality Configurations	1-9
Table 1-4	DG Board Options	1-10
Table 1-5	Standard DG5 Connectors with No Options	1-11
Table 1-6	DG5 GVO Connectors	1-11
Table 1-7	DG5 TVO Connectors.....	1-11
Table 1-8	DG5 DPLEX Connectors	1-11
Table 1-9	DG5-to-Monitor Cables and Adapters.....	1-20
Table 1-10	LEX/REX Hub Components and Uses.....	1-25
Table 1-11	Hub Cable Specifications	1-25
Table 1-12	PCI Audio Board Specifications.....	1-27
Table 1-13	DB-15 Pinouts for Video Sync and Digital Audio.....	1-29
Table 1-14	Speaker LEDs.....	1-30
Table 1-15	Digital Media XIO Options.....	1-31
Table 1-16	Basic L1 Functions	1-34
Table 1-17	Onyx 3000 Series Graphics Parts	1-35
Table 2-1	Onyx 3000 Series Configuration Overview	2-2
Table 2-2	Example Display Variables for 2 Graphics Seats, 2 Pipes Each .	2-9
Table 2-3	Display Variables and Pipe Numbers Associated with Example 2.....	2-10
Table 2-4	<i>Suggested</i> I/O-to-pipe Cabling: 1 I/O Rack in a Section	2-12
Table 2-5	<i>Suggested</i> I/O-to-pipe Cabling: 2 I/O Racks in a Section.....	2-13
Table 3-1	Starting and Stopping Graphics.....	3-2
Table 3-2	<i>ireeprom</i> Options	3-4
Table 3-3	Symptom and Cause Matrix.....	3-7

About this Booklet

This booklet provides quick-reference information on the **graphics components** of SGI Onyx 3000 series systems.

This booklet does not provide information on system components that are not graphics related. For information on nongraphics components, refer to *Hardware Components Quick Reference (SGI 3000 Family)*, publication number 108-0242-00x.

Note: Configuration information is subject to change due to new hardware or new manufacturing procedures.

This booklet is intended for experienced SGI service personnel.

Notational Conventions

Convention	Meaning
screen display	Fixed-space font denotes literal items such as references to commands, screen output, and examples.
<i>filename or variable</i>	Italic typeface denotes filenames, variable entries, document titles, and words that are being defined.
user input	Bold, fixed-space font denotes literal items that you enter on the keyboard in interactive sessions.

Supporting Documentation

The following table lists the reference documentation that supports the information in this booklet. These documents are available from <http://servinfo.csd.sgi.com>.

Supporting Document	Number
Hardware Components Quick Reference (SGI 3000 Family)	108-0242-00x
SGI Onyx 3000 Series Graphics System Hardware Owner's Guide	007-4264-00x
SGI Onyx 3000 Internal Technical Configuration Manual	108-0299-00x
SGI Onyx 3000 Family Configuration Rules	108-0310-00x
System Cabling (SGI 3000 Family)	108-0209-00x
System Components (SGI Origin 3000 Server Series)	108-0239-00x
System Controllers (SGI Origin 3000 Server Series)	108-0241-00x
Hardware Replacement Procedures (SGI Origin 3000 Server Series)	108-0249-00x
FRU and Tools Plan	Online only
Illustrated Parts Catalog	Online only
SGI X Window System Quick Reference Card	108-0253-00x
Digital Media Connections	007-3525-00x
Onyx2 Digital Video Multiplexer (DPLEX) Option Installation Guide	108-0192-00x
InfiniteReality Video Format Combiner User's Guide	007-3279-00x
DIVO Option and DIVO-DVC Option Owner's Guide	007-3524-00x

Onyx 3000 and G-brick Components

Chapter Contents

1.1	New Hardware Terminology	1-2
1.2	Standard Onyx 3000 Series Components	1-2
1.3	Onyx 3000 Series Component Block Diagrams	1-3
1.4	Onyx 3000 Series Rack Drawings	1-5
1.5	The G Brick	1-7
1.5.1	G Brick vs. KCAR (Onyx2)	1-7
1.5.2	G-brick Dimensions	1-7
1.5.3	G-brick Midplane	1-8
1.5.4	G-brick Components	1-8
1.5.5	G-brick-compatible Graphics Board Sets	1-9
1.6	DG5 Boards	1-10
1.6.1	DG5 Connector Overview	1-11
1.6.2	DG5-8 Notes	1-12
1.6.3	DG5-DPLEX Notes	1-12
1.6.4	DG5-DDO2 Notes	1-12
1.6.5	DG5-GVO Notes	1-12
1.6.6	DG5-TVO Notes	1-13
1.6.7	DG5 Board Drawings	1-13
1.6.8	DG5 Monitor Pinouts	1-16
1.6.9	Calculating DG5 Output Bandwidth	1-17
1.7	Ktown2 Board	1-17
1.8	RM Boards	1-18
1.9	GE16-4 Board	1-19
1.10	Onyx 3000 Series Graphics Displays	1-20
1.10.1	Monitor Cables	1-20
1.10.2	Onyx 3000 Series System with 1600SW Flat Panel Display	1-22
1.11	Keyboard, Mouse, and Hubs	1-23
1.11.1	The Standard Hub	1-23
1.11.2	The Extender Hubs	1-24
1.11.3	Mouse and Keyboard Controller Numbers (Device IDs)	1-26
1.11.4	Configuring Input Devices	1-27
1.12	Onyx 3000 Series Audio Components	1-27
1.12.1	Audio PCI Card	1-27
1.12.2	Speakers	1-29
1.13	SIO PCI Card	1-31
1.14	Digital Media XIO Options	1-31
1.14.1	DIVO Notes	1-31
1.14.2	DIVO-DVC Notes	1-32
1.14.3	XT-HD Notes	1-32
1.15	L1 Controller	1-33
1.15.1	L1 Hardware	1-33
1.15.2	Basic L1 Controller Functions	1-34
1.16	Part Numbers	1-35

1.1 New Hardware Terminology

G brick

Graphics insert module. The G brick is a modified KCAR (Onyx2) module that is compatible with Origin 3000 series systems, but not Onyx2 systems.

Ktown2

Modified Ktown board, which is not compatible with Onyx2.

Onyx 3000

Origin 3000 series system that contains G brick(s).

Onyx3

An unofficial term that has been misapplied to the Onyx 3000 series and to the G brick. Do not use this term.

1.2 Standard Onyx 3000 Series Components

All Onyx 3000 series systems contain the following major hardware components:

- One or more C bricks (with 2 or more processors each)
- One or more I bricks
- One or more G bricks (graphics modules); each G brick contains one or two graphics board sets (graphics pipes)
- Power bay(s) (However, graphics racks do not use power bays; the G-brick contains its own power supply.)
- An L2 system controller (The graphics rack does not contain an L2.)

1.3 Onyx 3000 Series Component Block Diagrams

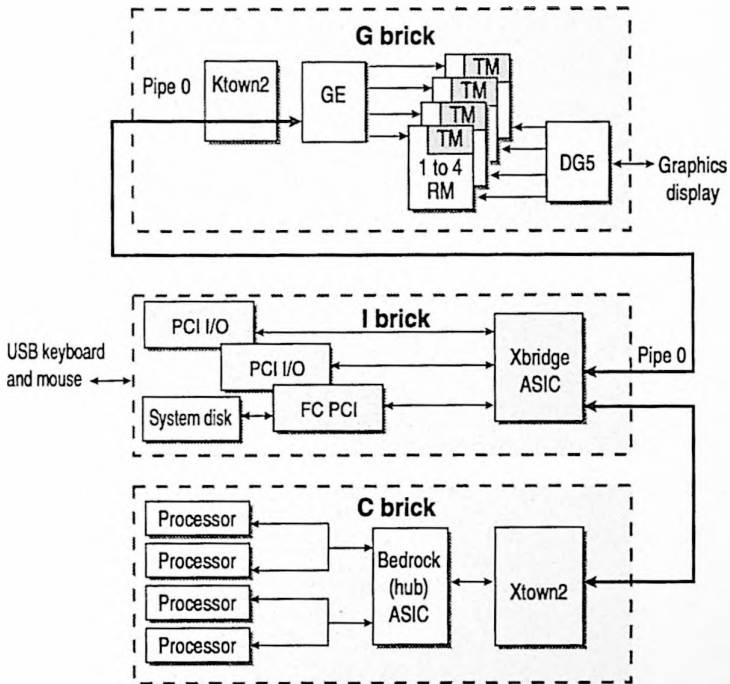


Figure 1-1 Single-rack Graphics Block Diagram Example

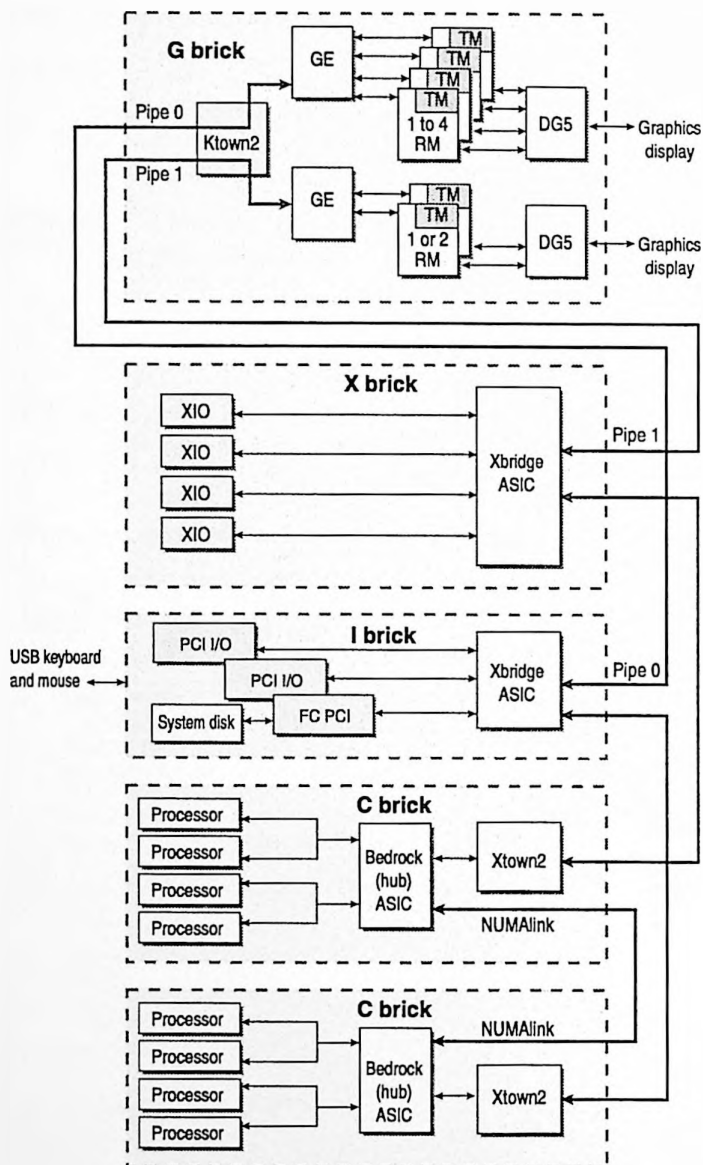


Figure 1-2 2-pipe Onyx 3000 Series System Block Diagram

1.4 Onyx 3000 Series Rack Drawings

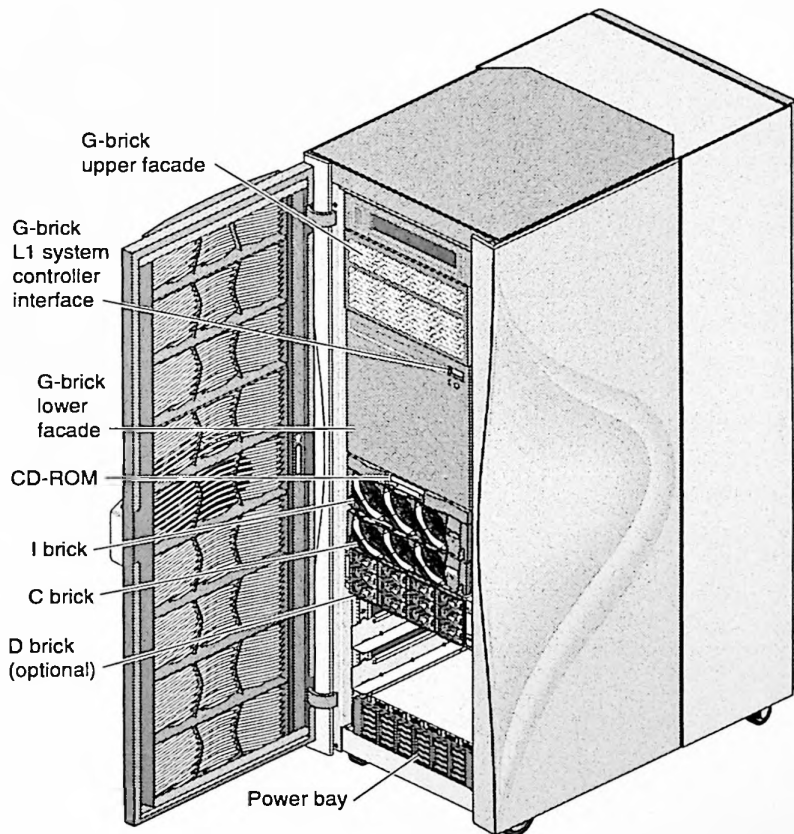


Figure 1-3 Single-rack System Components (Front View)

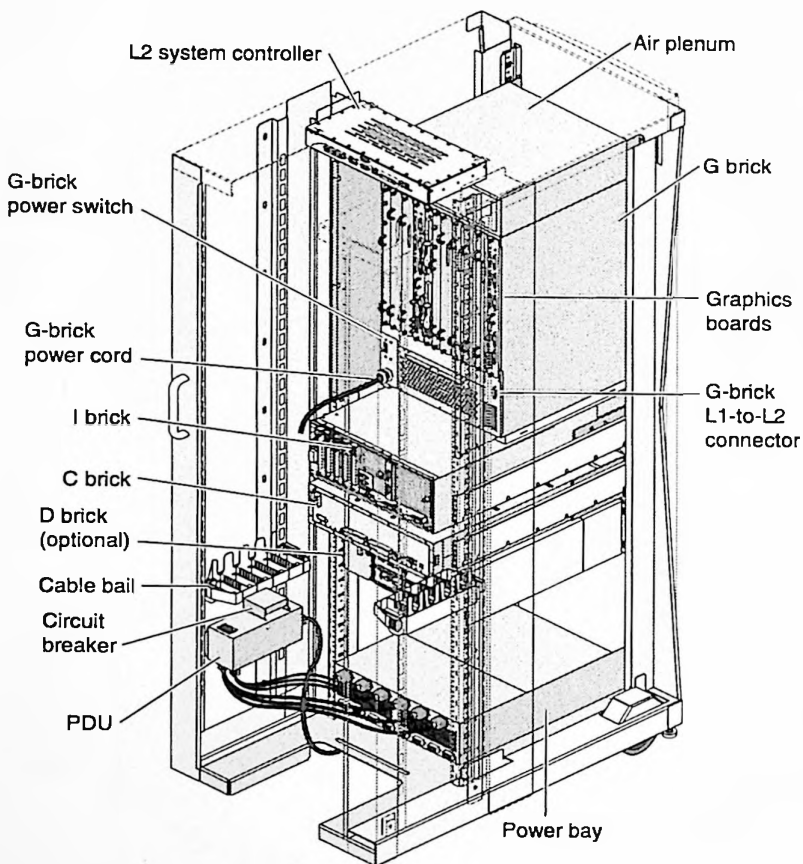


Figure 1-4 Major Components of the Single-rack System (Rear View)

1.5 The G Brick

1.5.1 G Brick vs. KCAR (Onyx2)

Table 1-1 Comparison: G Brick and KCAR

Description	G Brick	KCAR (Onyx2)
Host system	Origin 3000 series system	Origin 2000
Interface board	Ktown2	Ktown
Graphics-to-host processor connection	Xtown2 ASIC port on I/O bricks (I and X bricks)	Xtown ASIC port on Xtown XIO board
Graphics cable	Xtown2	Xtown
Pipes per brick/module	2 (1 each: 4RM & 2RM)	2 (1 each: 4RM & 2RM)
Board support	The G brick and KCAR both support InfiniteReality, InfiniteReality2, and InfiniteReality3 board sets.	
Midplane	The G brick and rackmount KCAR use the same midplane (013-1760-001).	
Module controller	L1	MSC
Rack controller	L2	MMSC
Power	The G brick and KCAR use the same components and require 208-Vac single-phase power.	
Cooling	The G brick and KCAR use the same cooling fans.	
Board slots	The G brick and KCAR use the same board slot locations.	
Primary keyboard & mouse connections	USB	PS2

1.5.2 G-brick Dimensions

Table 1-2 G-brick Technical Specifications

Specification	Measurement
Height	18U; 31.5 in. (80.1 cm)
Width	19.0 in. (48.3 cm)
Depth	20 in. (50.8 cm)
Weight (max.)	215 lbs. (97 kg)

1.5.3 G-brick Midplane

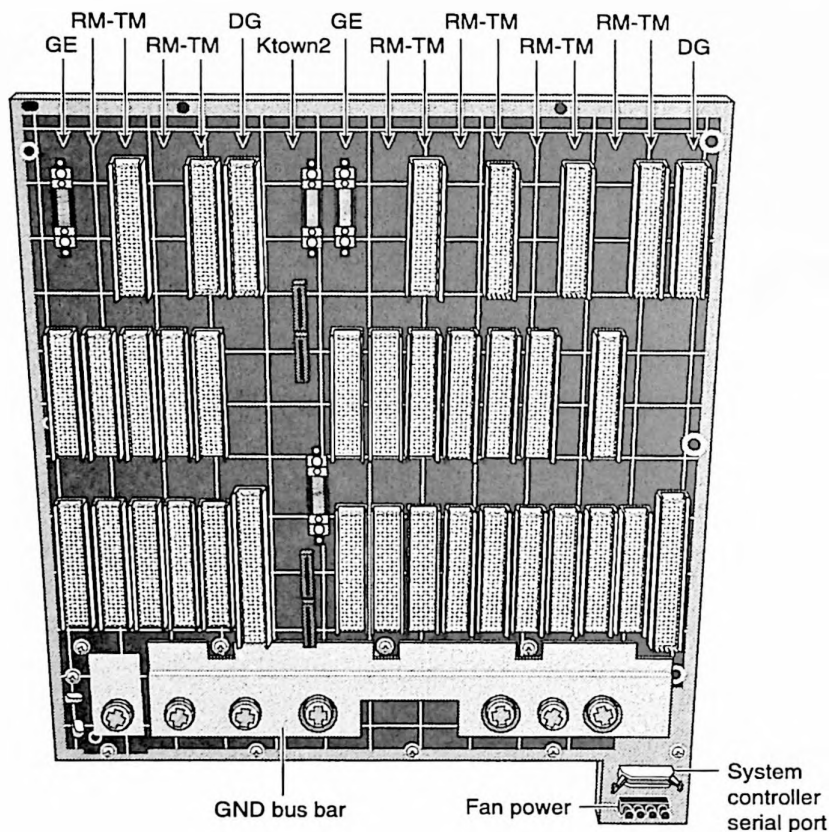


Figure 1-5 G-brick (and Onyx2 Rackmount) Midplane Rear View

1.5.4 G-brick Components

The G brick includes the following major components:

- GE board
- Ktown2 interface (supports two graphics pipes)
- G-brick L1 system controller
- 1, 2, or 4 RM/TM assembly boards per pipe
- 1 DG board per pipe (with or without an optional daughter card)

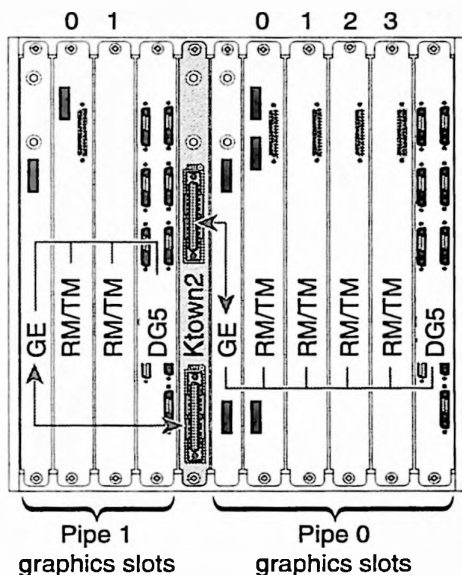


Figure 1-6 Graphics Pipe Board Locations

1.5.5 G-brick-compatible Graphics Board Sets

Table 1-3 InfiniteReality Configurations

Marketing Name	hinv output	GE board	RM board	Valid systems
InfiniteReality	InfiniteReality2	GE14-4	RM7	All Onyx2 & Onyx 3000 series
InfiniteReality2	InfiniteReality2	GE14-4	RM9	All Onyx2 & Onyx 3000 series
InfiniteReality2	InfiniteReality2E	GE16-4	RM9	All Onyx2 & Onyx 3000 series
InfiniteReality3	InfiniteReality3	GE14-4	RM10	All Onyx2 & Onyx 3000 series
InfiniteReality3	InfiniteReality3	GE16-4	RM10	All Onyx2 & Onyx 3000 series

Note: Pipes perform at the level of the lowest-performing hardware. For example, a pipe that includes one RM7 and three RM10s runs at the RM7 pace. However, SGI does not support the use of different RM boards within a pipe.

1.6 DG5 Boards

Table 1-4 lists the current DG5 boards that are compatible with Onyx 3000 series systems.

Table 1-4 DG Board Options

DG Board/ Option	Description
DG5-2	Basic configuration: 2 high-resolution RGB outputs.
DG5-8	8 high-resolution RGB output channels. Uses the VIO5H daughter card.
DG5-DPLEX	Digital (low voltage differential signaling [LVDS]) input and output for frame-multiplexed hyperpipe cascades, includes additional multiplex RGB output. Cascades the video outputs of two or more pipes to a single output.
DG5-DDO2	Digital output via DVP for connection to customer-designed hardware.
DG5-GVO	Digital dual-link (SDI) output (ITU-R 601) on two BNC connectors. Provides direct output from graphics to video in real time.
DG5-TVO	High-definition digital (transition minimized differential signaling [TMDS]) output for connection to DVI-compliant devices.

Note: All DG boards provide two RGB output channels and an analog composite video encoder. Each DG5 board supports only one option, such as GVO or TVO.

The DG5 board is always placed in the right-most slot in each graphics board set (pipe) of the G brick.

DG5 functions:

- Requests and receives digital frame buffer pixel data from the RM/TM board set.
- Handles all pixel clocking, genlocking, and cursor display functions.

A DG5 outputs a maximum of about 300 Mpixels/sec, which is split (often unevenly) among the output channels. You cannot configure channels that require a total output greater than 300 Mpixels/sec.

1.6.1 DG5 Connector Overview

Table 1-5 Standard DG5 Connectors with No Options

Connector	Type	Function
Monitors 0 through n	13W3	Variable high-resolution monitor outputs
S-Video	4-pin mini-DIN	Interface to S-Video VCR or monitor
CMPST 1	RCA jack	Interface to composite monitor or VCR
CMPST 2	BNC	Interface to composite monitor or VCR
Stereo View	9-pin D-sub	Interface to Stereo View device
Swap ready	BNC	Synchronizes front and rear framebuffer swapping between multiple pipes
Genlock in	BNC	Interface to house sync or other pipes (or video mixer)
Genlock loopthrough	BNC	Loop-through connection

Table 1-6 DG5 GVO Connectors

Label	Type	Function	Location
Link A	BNC (CCIR601)	Video output	Upper connector on right
Link B	BNC (CCIR601)	Video output	Lower connector on right

Table 1-7 DG5 TVO Connectors

Label	Type	Function	Location
TMDS-A	DVI-D	Video output	Upper connector on right
TMDS-B	DVI-D	Video output	Lower connector on right

Table 1-8 DG5 DPLEX Connectors

Label	Type	Function	Location
Digital Video In	LVDS	Digital video (LVDS) input	Lower connector on right
Digital Video Out	LVDS	Digital video (LVDS) output	Middle connector on right
Analog Video Out	13W3	DPLEX analog output	Upper connector on right

1.6.2 DG5-8 Notes

Channels 0 and 1 support *more bandwidth* than Channels 2 through 7. So, use Channels 0 and 1 for higher-resolution outputs, and use Channels 2 through 7 for lower resolution outputs. (Channels 0 and 1 have a 220-MHz maximum pixel clock rate, and Channels 2 through 7 have a 170-MHz maximum pixel clock rate.)

1.6.3 DG5-DPLEX Notes

- A hyperpipe cascade may contain from 2 to 8 pipes.
- Each pipe in a cascade must contain a DG5-DPLEX board.
- All pipes in a cascade must be cabled for genlock:
 - For internal sync, set the first pipe to internal loopback sync and set the other pipes to external sync.
 - For external (house) sync, set all pipes to external sync.
 - With 5 or more pipes, use a "star" video distribution amplifier (line driver or splitter) to distribute the genlock sync function from the source (house sync or primary pipe).
- All pipes in a cascade must be cabled for swap ready.

For cascades that contain more than 2 pipes, a BNC type T-connector is required for each additional pipe.
- LVDS cables connect the output of one pipe to the input of the next pipe.

1.6.4 DG5-DDO2 Notes

- Primarily used for custom hardware-in-the-loop (HITL) simulations.
- Digital output connects to customer-designed hardware (typically an LCD projection system).

1.6.5 DG5-GVO Notes

- Provides dual-link serial digital outputs (link A and link B)
 - 4:4:4:4 and 4:2:2:4 modes require both links.
 - 4:2:2 and CCIR 525/625 modes require link A only.
- Output does not support embedded digital audio.

- Supports two modes of operation:
 - Low-bandwidth, dependent mode (8 bits per pixel)
Outputs a portion of the selected monitor channel. A cursor appears in *both* the monitor channel and GVO outputs.
 - Low-latency, independent mode (10 bits per pixel)
Outputs directly from the frame buffer. The cursor appears in the GVO *or* monitor channel output. Requires external sync source (house sync).

1.6.6 DG5-TVO Notes

At printing, the TMDS output can be connected to the SGI video breakout box (VBOB) to provide SD or HD video output from the graphics pipe (HD-GVO solution).

In the future, additional TMDS connections should be supported.

1.6.7 DG5 Board Drawings

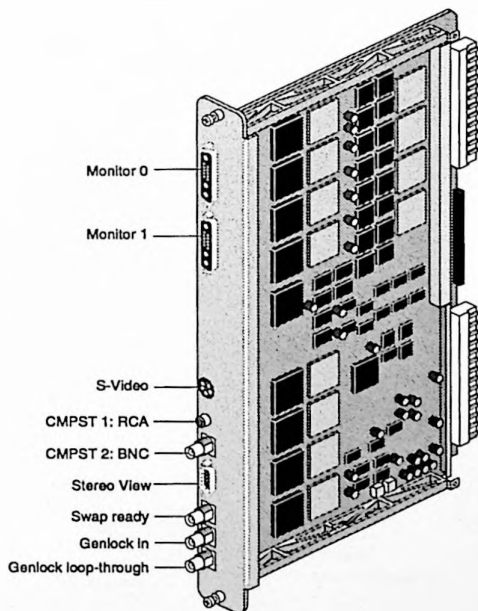


Figure 1-7 DG5-2 Graphics Board with No Options

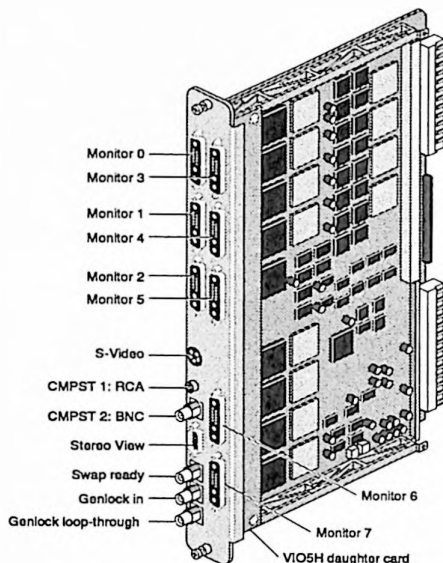


Figure 1-8 DG5-8 Board

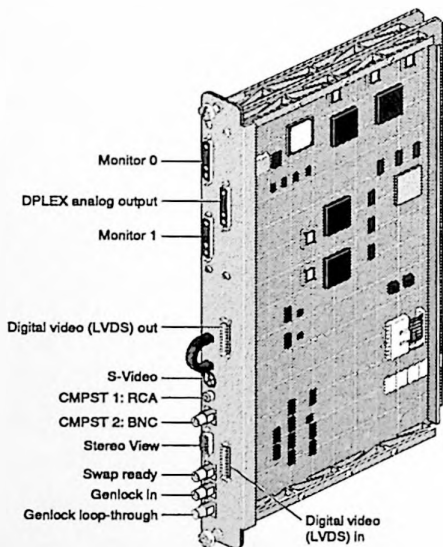


Figure 1-9 DG5-2 with Optional DPLEX Daughter Card

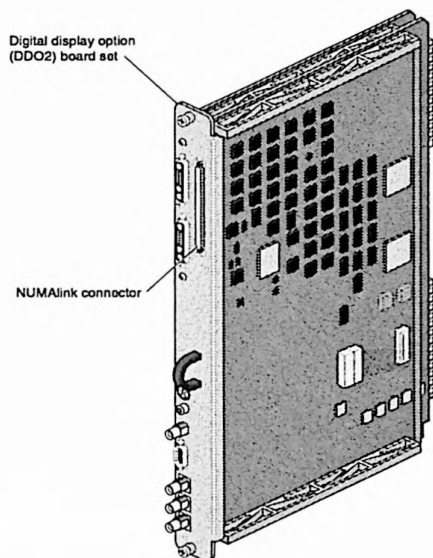


Figure 1-10 DG5-2 with Optional DDO2 Daughter Card

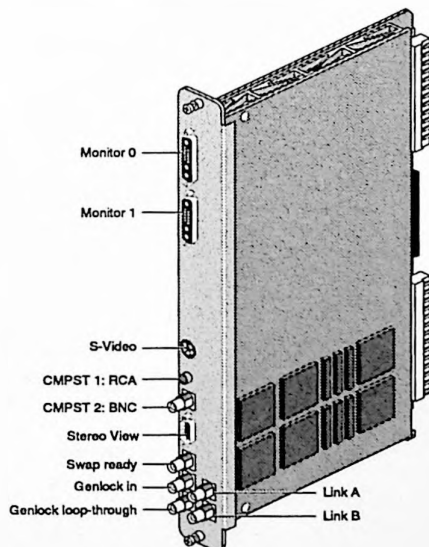


Figure 1-11 DG5-2 with Optional GVO Connectors

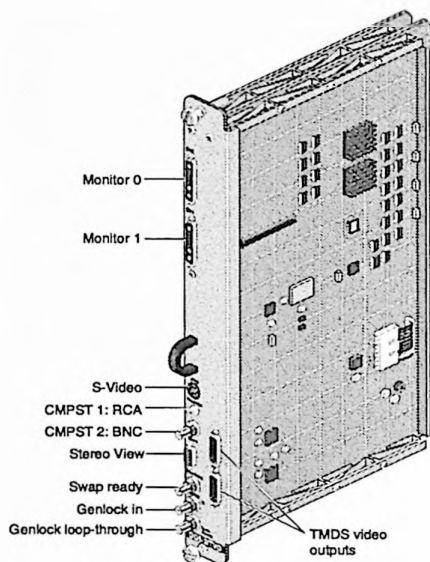


Figure 1-12 DG5-2 with Optional TVO Daughter Card

1.6.8 DG5 Monitor Pinouts

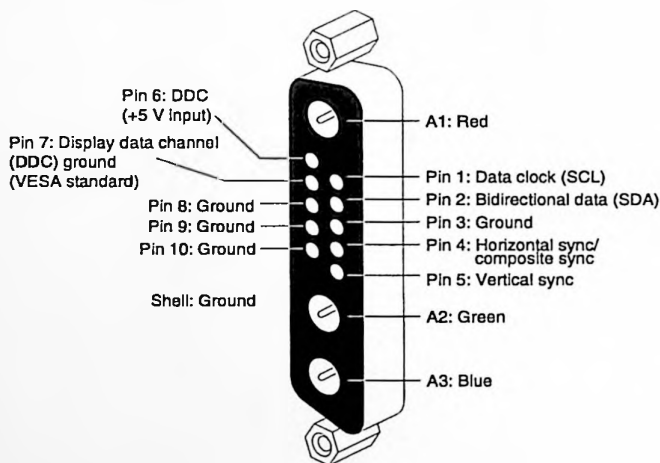


Figure 1-13 DG5 13W3 Monitor Pinouts

1.6.9 Calculating DG5 Output Bandwidth

Use the *ircombine* utility to calculate the required output bandwidth of a particular combination.

You can also use the following equations to estimate DG5 output bandwidth:

$$\frac{\text{pixels}}{\text{frame}} \times \frac{\text{frames}}{\text{second}} = \frac{\text{pixels}}{\text{second}}$$

$$\frac{\text{pixels/sec}}{1,000,000} = \text{MegaPixels/sec}$$

Example

$$\frac{(1024 \times 840) \text{ pixels}}{\text{frame}} \times \frac{60 \text{ frames}}{\text{second}} \times \frac{1 \text{ MPixel}}{1,000,000 \text{ pixels}} = \frac{51.6 \text{ MPixels}}{\text{second}}$$

1.7 Ktown2 Board

The Ktown2 board provides two (Xtown2 connections; the top connector is for the right pipe; the bottom connector is for the left pipe.

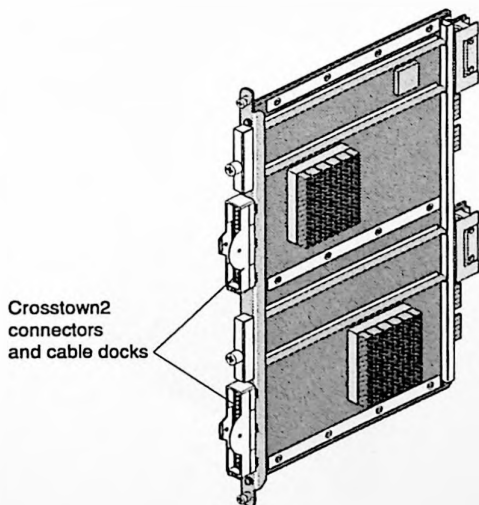


Figure 1-14 Ktown2 Board

1.8 RM Boards

RM boards provide buffer area for images. If you increase the buffer size (either by adding RM boards or by upgrading to RM boards with more memory), the hardware can store a greater number of images or store images with higher resolution.

A pipe with 4 RM slots requires 1, 2, or 4 RM boards; it will not function with 3 RM boards.

An RM board must always be in slot 0. Additional boards must be in adjacent slots. The following RM slot configurations are functional:

- 0
- 0, 1
- 0, 1, 2, 3

Refer to Table 1-3 on page 1-9 for a list of InfiniteReality RM boards.

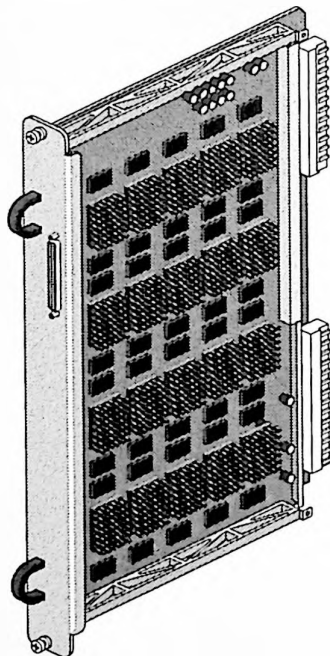


Figure 1-15 RM7 Board

1.9 GE16-4 Board

The GE16-4 board contains four processors that process OpenGL commands and vertex data that it receives from the host processors. Each pipe contains one GE16-4 board.

Refer to Table 1-3 on page 1-9 for a list of the GE boards associated with InfiniteReality hardware.

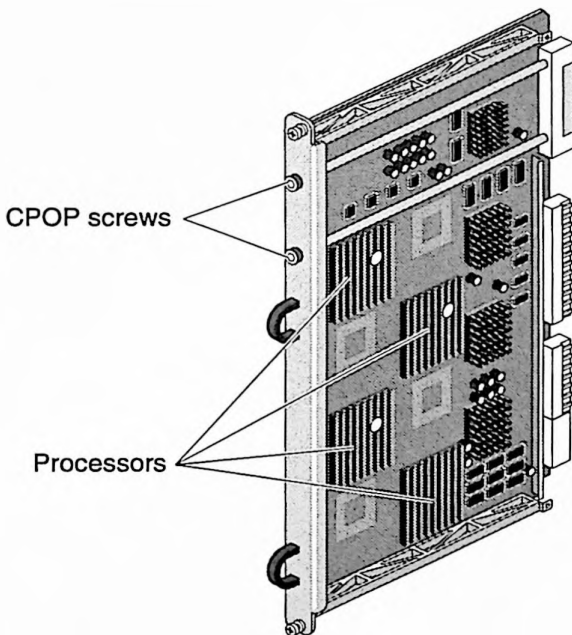


Figure 1-16 GE16-4 Board

1.10 Onyx 3000 Series Graphics Displays

- 24-inch monitor (standard)
 - 1920 x 1200 pixel resolution at 66 Hz
 - 16:10 aspect ratio
 - Connects directly to DG5 via 13W3 cable
- 21-inch monitor (option)
- 1600SW flat panel (option)
 - 1600 x 1024 pixel resolution
 - Requires MultiLink adapter (analog mode only)
 - Requires 3rd-party converter to view HDTV (720p format)

With a DG5-8, a graphics pipe supports up to 8 monitors.

1.10.1 Monitor Cables

Table 1-9 DG5-to-Monitor Cables and Adapters

Cable Type	Part Number	Additional Information
13W3-to-13W3	018-0646-002	Use with most monitors, such as 24-inch SuperWide
13W3F-to-13W3M with 2 BNC	018-0634-002	Has separate BNC connectors for horizontal and vertical sync
13W3-to-HD15	018-0839-001	Does not have sync pinouts for flat panel
13W3-to-HD15	018-0881-001	Has sync pinouts for flat panel
13W3F-to-5 BNCs	018-0633-001	Has separate connectors for R, G, B, horizontal sync, and vertical sync
DVI_D-DFP	9470466	Part of the flat-panel cable kit, 013-3066-001
VGA-to-DVI	9470467	Part of the flat-panel cable kit, 013-3066-001
DVI_D-DVI	9470468	Part of the flat-panel cable kit, 013-3066-001

SGI guarantees image integrity on monitors that have cables up to 30 ft. (9 m). When longer cables are required, SGI recommends that you use fiber-optic cables.

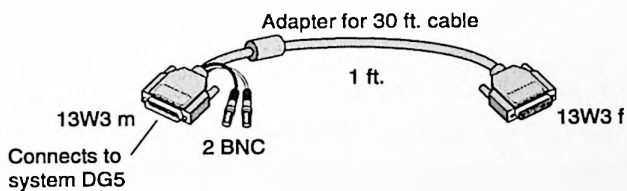
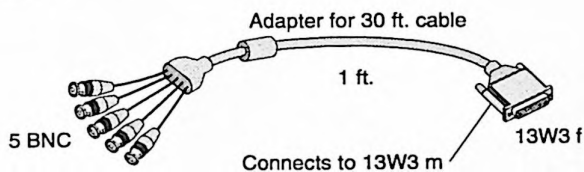
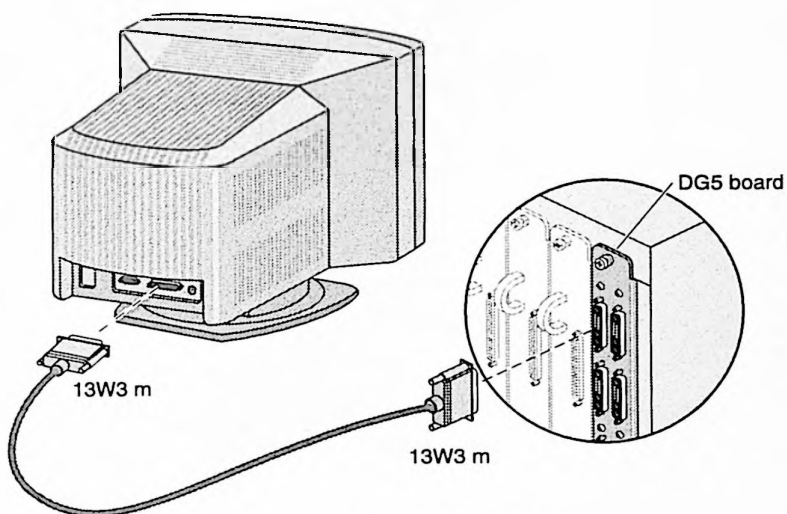


Figure 1-17 13W3 Monitor Cable and Adapters

1.10.2 Onyx 3000 Series System with 1600SW Flat Panel Display

To connect a Silicon Graphics 1600SW flat panel monitor to an Onyx 3000 series (or Onyx2) system, you need the following equipment:

- 1600SW with bundle (FPM00011), which includes
 - 1600SW monitor (061-0033-004)
 - Droid MultiLink adapter assembly (013-2840-001)
 - Assembly cable 3 pack (013-3066-001), which includes the VGA-to-DVI analog cable (9470467)
- 10-ft. 13W3-to-HD15 cable (018-0881-001)

Note: On other 13W3 cables, the H and V sync outputs may be on the wrong pins and, therefore, require rewiring.

- HD15 female-to-HD15 female adapter plug — or *Gender Bender* (not available through SGI)
- Special 1600x1024_60 video format file

Use the following procedure to configure the equipment:

1. Attach the 13W3 end of the 13W3-to-HD15 cable to DG5 Channel 0.
2. Attach the HD15 *Gender Bender* to the HD15 end of the 13W3-to-HD15 cable.
3. Use the VGA-to-DVI cable to connect the other end of the *Gender Bender* to the DVI connector on the MultiLink adapter.
4. Configure the MultiLink adapter for analog input (set the switch on the back of the unit to "A").
5. Connect the MultiLink adapter to the flat panel.
6. Use the *ircombine* utility to load the 1600x1024_60 video format.

Note: You will have to do this from a remote X display.

7. Configure the Channel 0 attributes as follows:
 - Turn off Sync on Green
 - Set Sync Output to "Horizontal"
8. Save the configuration to the GE EEPROM and restart graphics.

1.11 Keyboard, Mouse, and Hubs

The *hwgraph* supports up to 16 keyboard/mouse sets.

The keyboard and mouse set(s) connect to the USB ports on the I brick(s), either directly or via a hub.

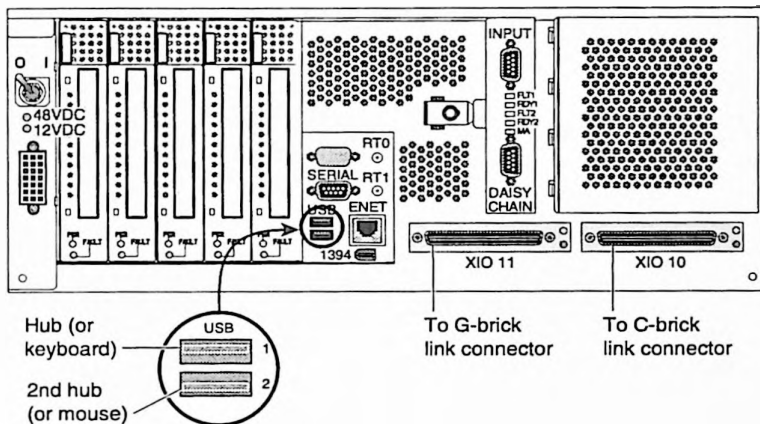


Figure 1-18 Primary I-brick Keyboard/Mouse/Hub Connectors

Two types of hub are available:

- The standard 4-port hub
- The USB extender hub set (LEX/REX)

1.11.1 The Standard Hub

The standard (single-piece) hub:

- Provides 4 USB A ports
- Uses the standard 3-m (9.84-ft.) USB cable
- Links keyboard/mouse sets (near the system) to an I brick
 - Connect to either USB port on an I brick
 - Use hub ports 1 and 2 for a keyboard/mouse set (any order)
 - Use hub ports 3 and 4 for a second keyboard/mouse set
- Links the G-brick L1 to the system L2 in multi-rack graphics systems (requires a dedicated hub), as shown on page 1-34
- Can be daisy chained by using any USB A port

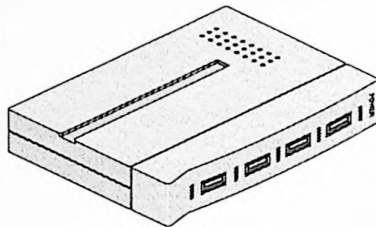


Figure 1-19 Standard Hub

1.11.2 The Extender Hubs

The USB extender hubs (LEX and REX):

- Extend the cabling distance for USB devices
- Link keyboard/mouse sets to an I brick
 - Connect LEX to I-brick USB port 1 or 2
 - Use REX ports 1 and 2 for a keyboard/mouse set (any order)
 - Use REX ports 3 and 4 for a second keyboard/mouse set
- May be daisy chained
- May connect other I/O devices such as a joystick
- Power on when they receive a signal from the I-brick at boot

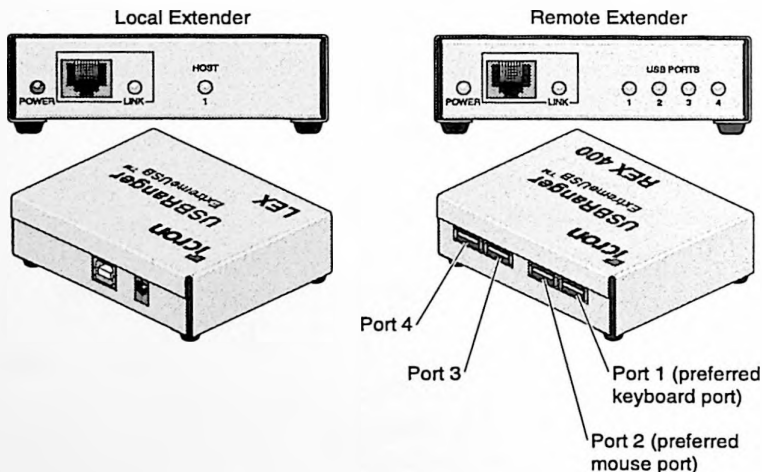


Figure 1-20 LEX and REX Hubs

Table 1-10 LEX/REX Hub Components and Uses

Hub Type	Ports	Use
Remote extender (REX)	4 USB ports and 1 differential port	Connects mouse and keyboard
Local extender (LEX)	1 USB port and 1 differential port	Connects to I-brick USB port; extends cabling distance

Table 1-11 Hub Cable Specifications

Connection	Maximum cable length	Connector types
I brick to LEX hub	3 m (9.84 ft.)	USB type A to USB B
LEX hub to REX hub	100 m (328 ft.) [9-m (30-ft.) cable shipped]	RJ45 to RJ45
REX to mouse/keyboard	3 m (9.84 ft.)	USB type A

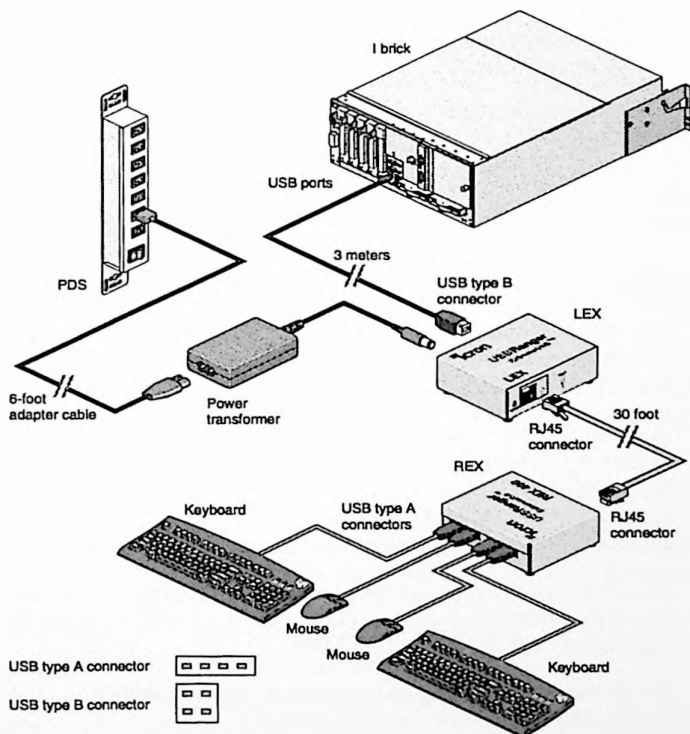


Figure 1-21 Standard Configuration for Extender, Hub, and Dual Seats

1.11.3 Mouse and Keyboard Controller Numbers (Device IDs)

ioconfig(1m) assigns controller numbers based on */etc/ioconfig.conf*.

At reboot, the system scans hardware. If a keyboard/mouse connection does not exist in */etc/ioconfig.conf*, the system assigns controller numbers (device IDs) to keyboard/mouse pairs and then writes these assignments to a new */etc/ioconfig.conf* file.

The assignment of controller numbers is determined by the NASIDs of the C bricks that connect to the I bricks where the keyboard/mouse pairs (and hubs if applicable) connect.

If you modify the hardware configuration, you can either:

- Edit */etc/ioconfig.conf* to reflect the changes. The changes take effect at the next reboot.
- Delete */etc/ioconfig.conf* and then reboot to rebuild it. (Note: Not recommended. It can change disk IDs and other IDs adversely.)

```
0 /hw/module/001c07/Ibrick/xtalk/15/pci/5/usb/1/1/keyboard
0 /hw/module/001c07/Ibrick/xtalk/15/pci/5/usb/1/2/mouse
```

The diagram illustrates the mapping of hardware paths to device IDs and their physical connections. It shows two paths from the hardware configuration: one for a keyboard (PCI slot 5, USB port 1) and one for a mouse (PCI slot 5, USB port 2). These paths converge at a single point labeled 'USB controller (in PCI slot 5 of system I brick)'. From this controller, two lines branch out: one to 'I-brick port (USB port 1)' and another to 'Port on USB hub (keyboard on 1; mouse on 2)'. The 'I-brick port' is associated with 'Device ID (controller 0)'.

Figure 1-22 Example Device IDs and Paths in */etc/ioconfig.conf*

The controller number determines the keyboard/mouse sets */dev/input* entry. When the X Window System server starts, the *-devdir* option controls where to look for input devices; by default, */dev/input* is used.

The first X server (and the keyboard/mouse set assigned as controller 0) uses the */dev/input* directory. Additional X servers (and keyboard/mouse sets) use */dev/input#*, where # is the controller number assigned to the keyboard/mouse set in the *ioconfig.conf* file. For example, the second X server uses the */dev/input1* directory.

Device IDs are unique only among devices of a given type. For example, controller number 0 may be assigned to several devices — as long as they are unique device types.

Check the */dev/input* files to ensure that the keyboard and mouse that are used by an X server have the same ID.

Use the *hinv* command to display device IDs.

1.11.4 Configuring Input Devices

The system does not configure a mouse or keyboard unless it is connected prior to boot, or you restart graphics, or you perform the following procedure to reconfigure the keyboard/mouse setup:

1. Stop graphics. For example, run `/usr/gfx/stopgfx`.
2. Add or move devices as necessary.
3. Run `/sbin/ioconfig -f /hw` to assign temporary device IDs and to modify `/etc/ioconfig.conf`.
4. Check the device IDs in `/etc/ioconfig.conf`. If needed, edit `ioconfig.conf` to assign the correct IDs.
5. If needed, edit the `Xservers` file.
6. Rerun `/sbin/ioconfig -f /hw` if you made changes in Step 4.
7. Restart graphics. For example: run `/usr/gfx/startgfx`.

Note: Never disconnect input devices while booting UNIX because it may cause an immediate or delayed system crash.

1.12 Onyx 3000 Series Audio Components

1.12.1 Audio PCI Card

For system audio output, Onyx 3000 series systems use a PCI card (030-1649-001) installed in the system I brick.

Warning: Do not use first or second generation audio boards (P/N 030-0950-00x) with Onyx 3000 or Origin 3000 series systems. System damage or fire will result.

Table 1-12 PCI Audio Board Specifications

Feature	Specification
Synchronization	Locks to professional video (via video sync) and audio equipment with professional jitter attenuation. Accepts video reference input, generates AES11 grade 2 (10 ppm) clocks.
Input sample rates	Continuous between 32 kHz and 48 kHz.
Output sample rates	32 kHz, 44.1 kHz, and 48 kHz.
Jitter attenuation	For 44.1-kHz or 48-kHz sample rates.
Video reference loopthrough	75-ohm BNC connectors, breakout cable.

Table 1-12 PCI Audio Board Specifications (continued)

Feature	Specification
Coaxial digital audio input and outputs	AES-3id 75-ohm BNC connector, breakout cable. AES11 synchronization input (for audio clock rates). AES3 professional 2-channel 24-bit digital. Compatible with IEC 958, S/PDIF consumer 2-channel digital.
Optical digital input and output	12.8-Mbps SHARP multimode plastic fiber-optic connector, PCI I/O panel. 8-channel, 24-bit ADAT optical interface. Compatible with IEC 958, S/PDIF consumer 2-channel digital.

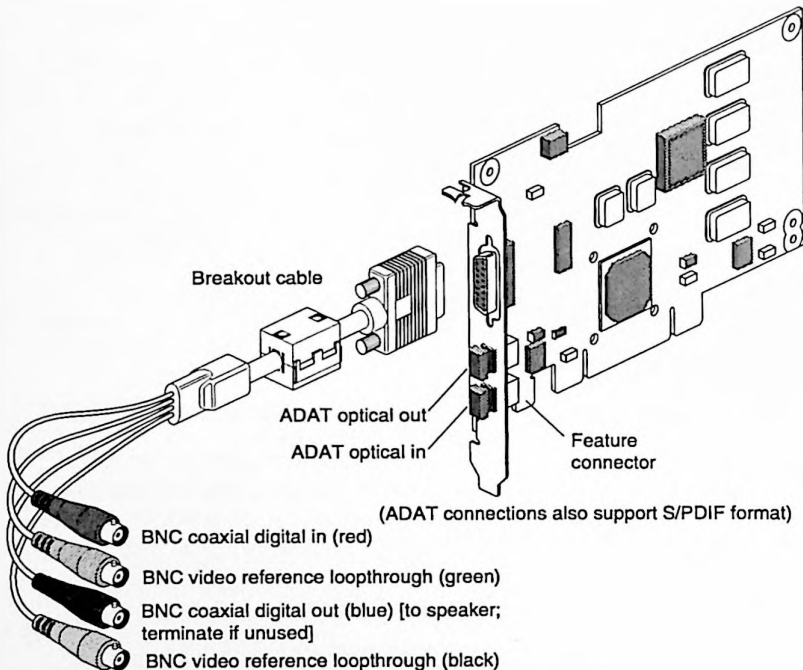


Figure 1-23 Audio PCI Card and Breakout Cables

Table 1-13 DB-15 Pinouts for Video Sync and Digital Audio

Pin	Assignment	Pin	Assignment
1	COAXIAL DIGITAL_IN GND	9	COAXIAL DIGITAL_OUT
2	COAXIAL DIGITAL_IN	10	NOT CONNECTED
3	VID_REF	11	COAXIAL DIGITAL_OUT GND
4	VID_REF GND	12	NOT CONNECTED
5	VID_REF	13	CHASSIS GND
6	VID_REF GND	14	CHASSIS GND
7	NOT CONNECTED	15	NOT CONNECTED
8	CHASSIS GND		

1.12.2 Speakers

Onyx 3000 series systems use Genelec 2029B digital speakers.

In digital mode, the volume control on the master speaker controls both speakers. Set the slave to maximum.

To avoid fuse or speaker damage, set each speaker's 110/220-Vac Voltage switch to the proper voltage (default is 220 Vac).

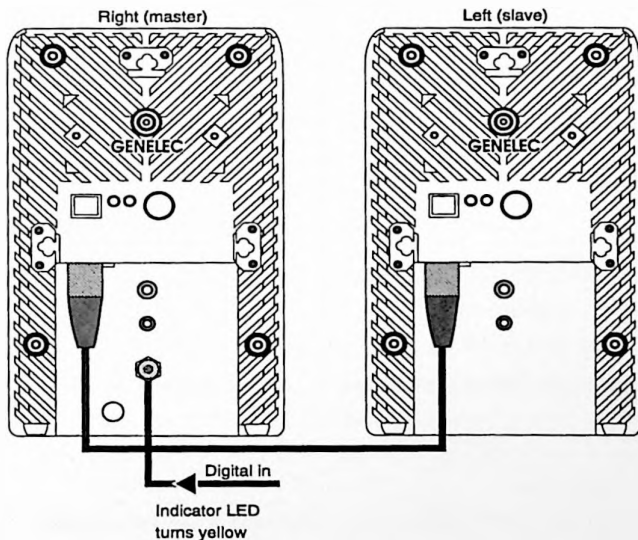


Figure 1-24 Speaker Cabling

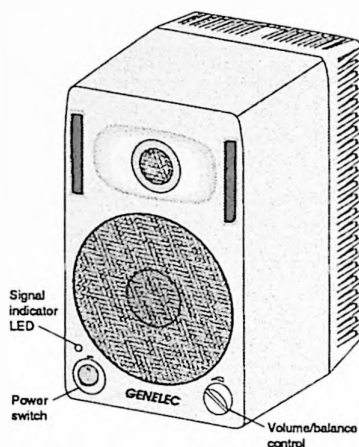


Figure 1-25 Speaker Front

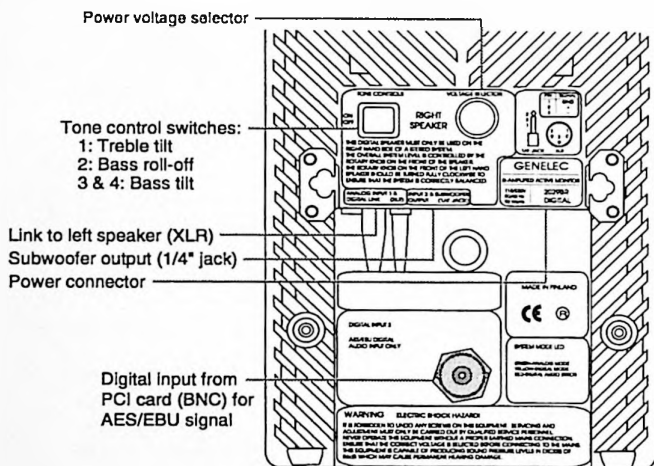


Figure 1-26 Rear Controls and Connectors (Right Speaker)

Table 1-14 Speaker LEDs

LED	Description
Green	Power on with no digital input (normal for analog)
Yellow	Power on with active digital input carrier detected (normal for digital)
Red	Bit errors detected in digital transmission

1.13 SIO PCI Card

- Connects to RS-232 or RS-422 serial devices like joysticks
- Is currently being qualified

1.14 Digital Media XIO Options

Table 1-15 Digital Media XIO Options

XIO Card	Description
DIVO	Provides broadcast-quality digital video and audio I/O for Onyx 3000 series, Onyx2, and Origin systems. The DIVO option supports: - YCrCb and RGB in 4:4:4 formats - YCrCb in 4:2:2:4 format - ITU-R 601 525/625 timings - 24-bit digital audio in AES/EBU and ADAT formats Origin systems that are equipped with DIVO can function as video "servers." Onyx2 and Onyx 3000 series systems may combine DIVO with DG5-GVO for adding graphics to a video input in real-time.
DIVO-DVC	Same features as DIVO but includes support for the DVCPRO video format.
XT-HD	Provides real-time, uncompressed, high-definition video I/O for Onyx 3000 series, Onyx2, and Origin systems. The XT-HD option supports: - YCrCb and RGB in 8- or 10-bit format - 4:2:2 (:2) and 4:4:4 (:4) samplings

1.14.1 DIVO Notes

- Use the video library control panel (*/usr/sbin/vcp*) to configure the DIVO source (input) and drain (output)
- Commands are located in */usr/dmedia/bin/DIVO/*
 - *divo_ee* - end-to-end loopthrough (source-to-drain)
 - *divo_memtovid* - directs specified file to output
 - *divo_vidtomem* - captures input and stores it to a file
 - *flashdivo* and *flashalldivo* - updates the built-in board software
 - *divo_reset* - reinitializes the DIVO board
 - *rgbtorice* and *ricetorgb* - provides Rice compression and decompression for RGB files
- Diagnostics are located in */usr/diags/DIVO/bin/*
 - *divo_confidence* - full confidence test for DIVO board
 - *loopback* - provides loopback testing

1.14.2 DIVO-DVC Notes

- DIVO-DVC is similar to DIVO but includes support for the DVCPRO video format.
- Unlike the DIVO board, the DIVO-DVC does not have on-board jumpers for configuring the GPI ports. The DIVO-DVC GPI ports support only switch-closure mode.

1.14.3 XT-HD Notes

- To test the XT-HD board, disconnect the 200-pin cable from the board and enter:
`/usr/diags/XTHD/bin/xthd_confidence`
- Two breakout cable types are supplied with the XT-HD board. Both cable types consist of one 200-pin connector and four 50-pin connectors. The 50-pin connectors differ for each cable type to support:
 - Panasonic equipment (such as the D-5 mastering VTR)
or
 - Philips equipment (such as the Spirit DataCine)

Note: These cables provide parallel digital I/O connections. Most HDTV equipment requires serial I/O connections (75-ohm coaxial cable with BNC connectors). Third-party parallel/serial converters are commonly used to provide compatibility.

- The XT-HD board supports both analog (genlock) and tri-level sync sources.
- Use the video library control panel tool (`/usr/sbin/vcp`) to configure the XT-HD source (input) and drain (output)

1.15 L1 Controller

1.15.1 L1 Hardware

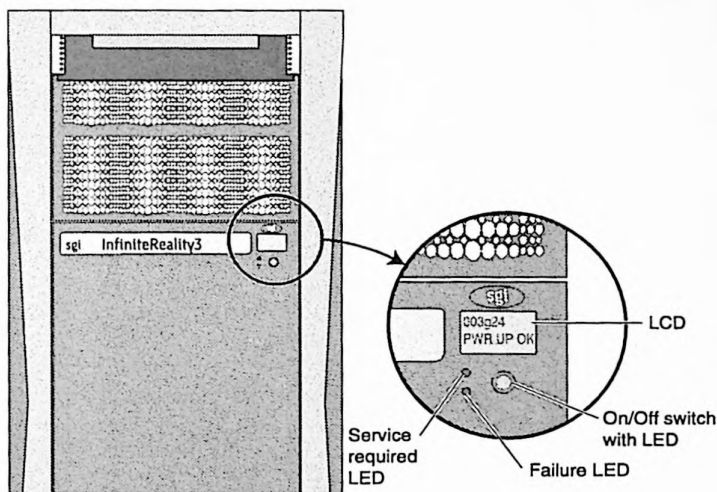


Figure 1-27 G-brick L1 Display, LEDs, and Power Button

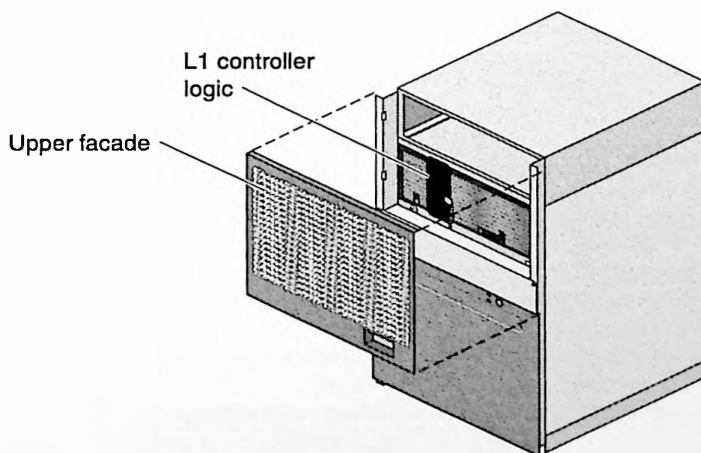


Figure 1-28 Location of L1 Controller Logic

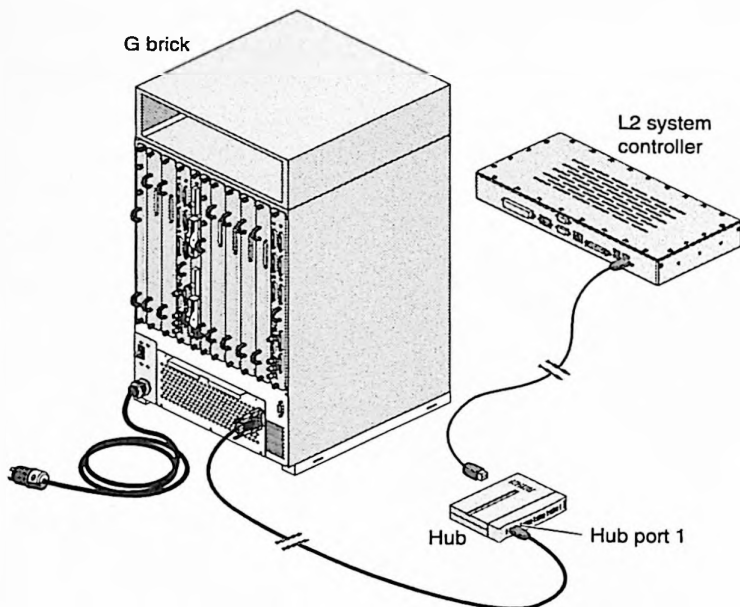


Figure 1-29 G-brick L1 to System L2 Cabling Example

Note: Do not "hot plug" or remove cables while the system is running. This can hang or crash the entire system.

1.15.2 Basic L1 Controller Functions

Table 1-16 Basic L1 Functions

L1 Function	Used in G brick?
Reports voltage increases, decreases, or failures	Yes
Controls and monitors fan speed	Yes
Monitors and controls LEDs	Yes
Monitors the power On/Off switch	Yes
Controls voltage regulator modules (VRMs)	No
Controls voltage margins within the brick	No
Reports operating temperature and status of 48-Vdc input power	No
Reads system identification (ID) PROMs	No
Monitors the reset switch and nonmaskable interrupt (NMI) switch	No

For L1 commands, refer to Chapter 4 of the *Hardware Components Quick Reference (SGI 3000 Family)* booklet, 108-0242-00x.

1.16 Part Numbers

Table 1-17 Onyx 3000 Series Graphics Parts

Part Number	Description	Common Name
013-1677-001	PCA Set RM7 & TM7-64	RM7
013-1679-001	PCA DG5-8 & VIO5H	DG5-8
013-1740-001	Blower Assy KCAR	Fan
013-1745-001	Baffle	RM slot baffle
013-1760-001	Midplane Assy with Stiffeners	Midplane
013-1809-001	Power Entry Assy US KCAR	G-brick power entry
013-1972-002	PCA Set DG5-2 & GVO	DG5-2/GVO
013-2447-002	Assy PCA Set DG5-2 & DDO3	DPLEX; DG-5/DD03
013-2464-001	PCA SET RM9 & TM7-64	RM9
013-2614-001	Carrier Assy PCI	PCI sled
013-2688-002	DG5-2/TVO Board Set	DG5-2/TVO
013-2746-002	VBOB System Assy w/o TIO Card	Video breakout box, no TOI
013-2840-001	Droid Assy (Universal Interface Adapter)	MultiLink adapter; Droid
013-2947-002	KCAR L1 Controller	G-brick L1
013-2954-001	PCA Set RM9 and TM8-256 (RM-10)	RM10
013-3066-001	Cable Assy 3-pack	Cables for flat panel
013-3005-001	PCA DG5B-8 & VIO5H	DG5B-8 upgrade
013-3100-001	Cable Assy Diff DNET 1M Red	Xtown2n cable (red)
013-3101-001	Cable Assy Diff DNET 2M Grn	Xtown2 cable (green)
013-3102-001	Cable Assy Diff DNET 3M Yel	Xtown2 cable (yellow)
013-3103-001	Cable Assy Diff DNET 4M Blu	Xtown2 cable (blue)
013-3229-001	Plenum Assy Intake Onyx 3	G-brick inlet air duct
015-0261-001	Harness Assy AC US KCAR	U.S. G-brick power cord
013-1914-001	Assy Power Entry KCAR Intl	Internatl G-brick power cord
015-0347-001	Harness Assy KCAR L1 Display	L1 display cable
018-0633-001	Cable Assy Adpt 1Ft 13W3 F to 5 BNC	13W3 F to 5 BNC
018-0634-002	Cable Assy Adpt 13W3M to 13W3F 2BNC	13W3-to-13W3, BNC sync
018-0646-002	Cable Assy 13w3 to 13w3 male 30ft	13W3-to-13W3 cable
018-0648-001	Cable Assy Radical Breakout PCI	Audio breakout cable
018-0695-001	Cable Assy ADAT Optical 2M	ADAT cable for audio PCI
018-0731-001	Cable Assy Radical PCI Digital Sync	Sync cable
018-0792-001	Cable Assy DPLEX/Nclops [40-pin]	1.5-m LVDS digital cable
018-0809-001	Cable Assy USBa to USBb 4Pos 3M Blk	USB cable
018-0831-001	Cable Assy Cable SMB-PC to BNC	Internal VBOB cable
018-0839-001	Cable Assy ADAP 13W3 to HD15 1'	13W3 to HD15 cable
018-0859-003	Cable Assy KCAR L1 DB9 USB L2	L1 to L2-cabled hub cable
018-0881-001	Cable Assy Video 13W3 to HD15 10Ft	13W3-to-HD15 for flat panel
018-0911-001	Cable Assy RJ45 to RJ45 Cat5 30ft	RJ45 cable
018-0925-001	Cable Assy BNC to BNC RG59 30ft	BNC cable
018-0942-001	Cable Assy USBa to B 4Pos 10ft	USB cable (port type A to B)

Table 1-17 Onyx 3000 Series Graphics Parts (continued)

Part Number	Description	Common Name
018-0948-001	Cable Assy DVI_D-DVI_D 1 Link 5M Bl	Black TMDS cable
018-0973-001	Cable Assy DVI_D-DVI_D 1 Link 5MWh	White TMDS cable
018-0987-001	Cable Assy BNC TO BNC RG59 16.405Ft	BNC timing reference cable
024-1440-001	Label Magnetic Onyx3 IR3	InfiniteReality3 label
024-1521-001	Label Magnetic IR	InfiniteReality label
024-1522-001	Label Magnetic IR2	InfiniteReality2 label
030-1051-002	PCA GE14-2	GE14-2 board
030-1055-001	PCA DG5-2	DG5-2 board
030-1129-002	PCA GE14-4	GE14-4 board
030-1398-001	PCA GE16-4	GE16-4 board
030-1439-002	PCA L1 System Control Display	L1 display
030-1537-001	TOI Card	TOI option for VBOB
030-1593-001	PCA Assy Ktown2	Ktown2 board
030-1649-001	PCA Radical Audio PCI	Audio card
030-1657-001	PCA PCI SIO UFC	SIO card
040-2996-001	VBOB Bracket	VBOB mounting ears
040-3006-001	Bracket Front Left Onyx3 Rack	G-brick mounting bracket
040-3007-001	Bracket Front Right Onyx3 Rack	G-brick mounting bracket
040-3008-001	Bracket Rear Left Onyx3 Rack	G-brick mounting bracket
040-3009-001	Bracket Rear Right Onyx3 Rack	G-brick mounting bracket
040-3190-002	Bracket Mounting Onyx3 L1 Display	G-brick L1 display bracket
040-3192-001	Panel Facade Top Onyx3	G-brick top trim/skin
040-3193-001	Panel Facade Bottom Onyx3	G-brick bottom trim/skin
040-3201-001	Shelf Module	Utility shelf module
042-0847-001	Speaker Genelec 2029B Digital	Digital speakers
061-0021-001	Monitor 24" 30 96KHZ	24-in. SuperWide monitor
060-0026-002	Power Supply AC/DC 3100WDC	G-brick power supply
061-0033-004	Monitor FP 17.3In Model 1600SW W/SE	Flat panel monitor
061-0036-001	Monitor 24 Inch So-Hem Sony 1920X1200	SuperWide monitor, S. Hemi.
061-0043-001	Monitor 21" G1 Northern Hemisphere	21-inch monitor, N. Hemi.
061-0045-001	Flat Panel 17.3" 1600SW w/Blue Bezel	Flat panel monitor (blue)
062-0066-001	Keyboard USB USA w/beep	USB keyboard
063-0011-001	Mouse 3 Button USB	USB mouse
9350102	Pwr Cord 125V 10A 18AWG 8.2FT Dom	VBOB power cord - USA
9350848	Pwr Cord 250V 2.5A IEC320 C14-C7 6F	LEX/REX hub power cord
9470466	Cbl 2M Blk Sgl Lnk DVI_D-DFP Dig w/Fer	Flat panel, droid cable
9470467	Cable Assy 2M Blk DVI_A-VGA w/Fer	VGA-to-DVI analog cable
9470468	Cbl 2M Blk Sgl Lnk DVI_D-DVI w/Fer	Flat panel, droid cable
9470545	Hub USB 4-Port 100-240VAC	Standard hub (not LEX or REX)
9470546	Hub XTNDR USB-CAT5-USB 4Port	USB extender hub (LEX/REX)
9660100	Termn BNC 75 Ohm No Chain	75-ohm BNC terminator

Onyx 3000 Series Configuration

Chapter Contents

2.1	Configuration Overview	2-2
2.2	Basic Configuration Rules	2-2
2.3	Example Rackmount Graphics Configurations	2-3
2.4	Graphics Rack Layout and Bay Numbering	2-4
2.5	Graphics Rack Placement	2-5
2.6	Assigning Pipe Numbers	2-6
2.7	Suggested Pipe Configuration	2-7
2.8	Checking Actual Pipe Numbering	2-9
2.9	Pipe and X Server Numbering	2-9
2.9.1	hostname:0.0 Described	2-9
2.9.2	The Xservers File: Configuring Graphics Seats and Assigning Keyboard and Mouse Control	2-10
2.9.3	Pipe Failure and Pipe Numbering	2-11
2.10	Recommended I/O-bricks-to-G-brick Connections	2-12
2.11	Upgrade Notes	2-13
2.12	Power Cabling	2-14
2.13	Keyboard and Mouse Configuration	2-14

2.1 Configuration Overview

Table 2-1 Onyx 3000 Series Configuration Overview

	3200	3400	3800
No. of processors	2 to 8	4 to 32	16 to 512 (with 4P upgrades)
No. of pipes	1 or 2	1 to 8	1 to 16
No. of racks	1	2 to 6 (1 to 4 graphics racks)	3 to 32 (4 to 8 graphics racks)
Marketing code of base product	ONYX-3200	ONYX-3400	ONYX-3800

Note: At printing, systems with more than 32 processors or 9 pipes had not been qualified.

2.2 Basic Configuration Rules

- Pipe 0 must connect to the system I brick.
- Each pipe must connect (via an I/O brick) to a C brick.
In theory, a 4-processor C brick can support 2 pipes (2 processors per pipe). However, due to current limitations of Xtown2 connections on I/O bricks, each pipe requires a dedicated C brick.
- Each pipe requires an Xtown2 connection.
 - Each pipe requires one I brick or one X brick.
 - The Xtown2 ports that connect to pipes should be set to 400 MHz (default setting on production I and X bricks).
 - Do not connect a pipe to a P brick. (P-brick Xtown2 ports run at 600 MHz rather than 400 MHz, as required for graphics.)
- Xtown XIO boards are not required for graphics connections in X bricks.
- RM slot 0 (the slot closest to the GE board) must be populated.
- Install RM boards in adjacent slots. Do not leave empty slots between RM boards.
- Pipes require 1, 2, or 4 RM boards; they do not function with 3 RM boards.

2.3 Example Rackmount Graphics Configurations

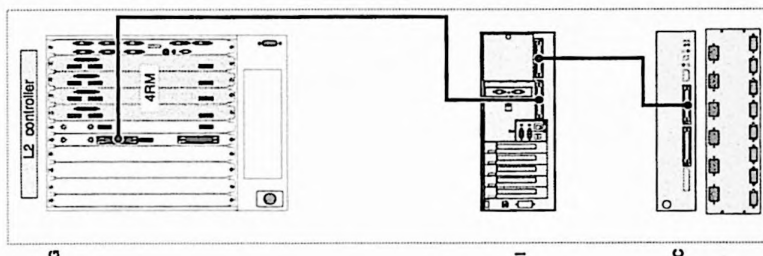


Figure 2-1 Single-rack, Single-pipe Configuration Example

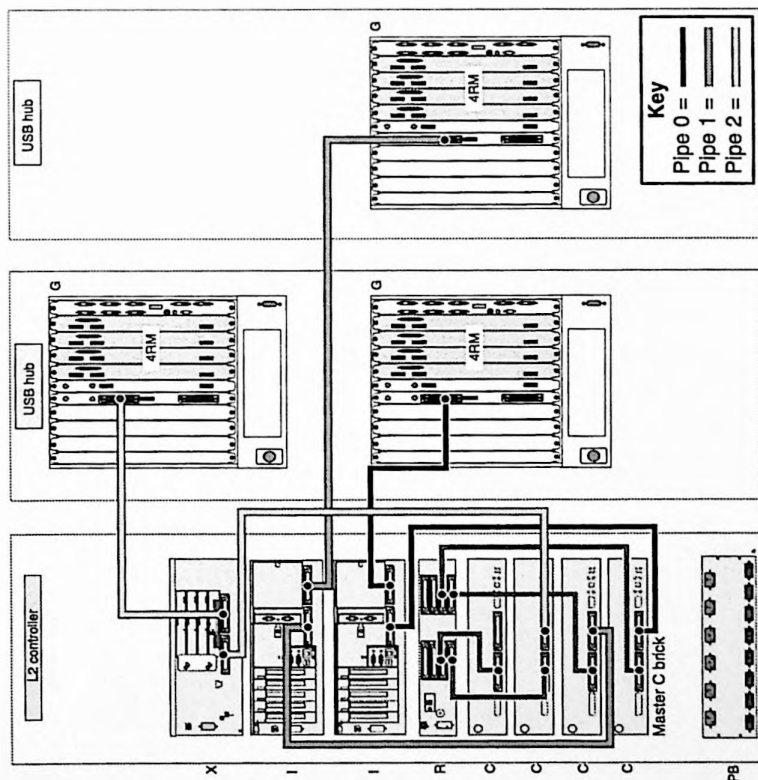
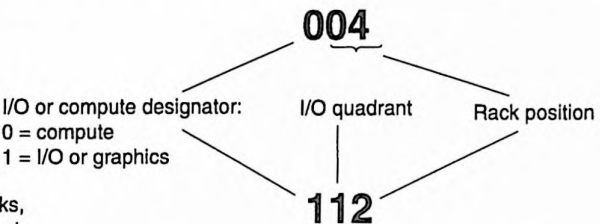
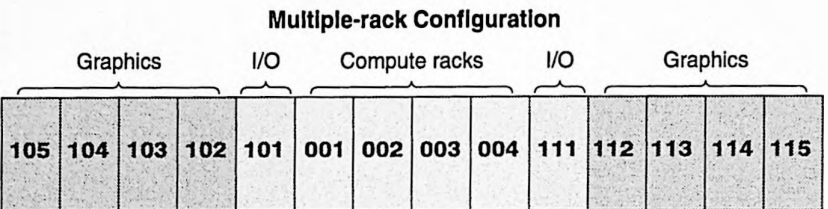


Figure 2-2 Three-rack, Three-pipe (4RM) Configuration Example



In graphics racks,
install G bricks at
bays U01 and U20.

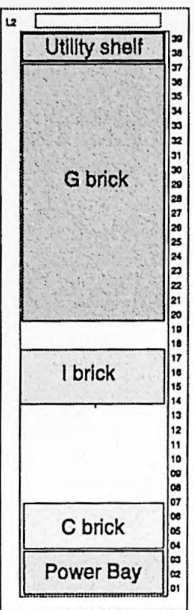


Figure 2-3 Single-rack Configuration

Bay (Slot) Numbering and Rack Layout

2.5 Graphics Rack Placement

One I/O rack:

Add graphics racks to the left of the I/O rack as you face the system front.

Two or more I/O racks:

Add graphics racks alternately on the left and right sides of the system.

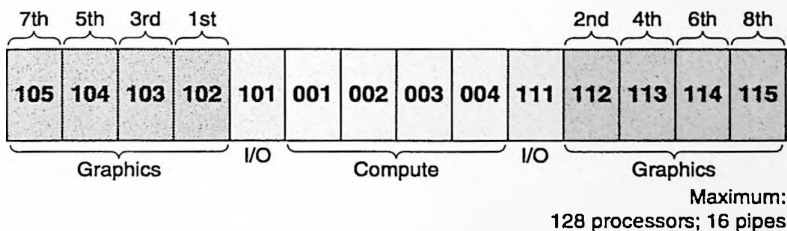
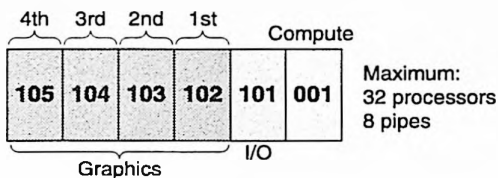
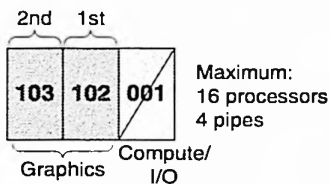


Figure 2-4 Onyx 3000 Series Rack Placement

2.6 Assigning Pipe Numbers

- The system automatically assigns pipe numbers.
- Pipe numbers are based on the NASIDs of the connected C bricks.

Pipes are assigned sequentially from the C brick with NASID 0 to the C brick with the highest NASID.

Example

C brick	NASID	Pipe Number
001c07	0	0
001c10	1	1
001c13	2	2
001c16	3	3

connects

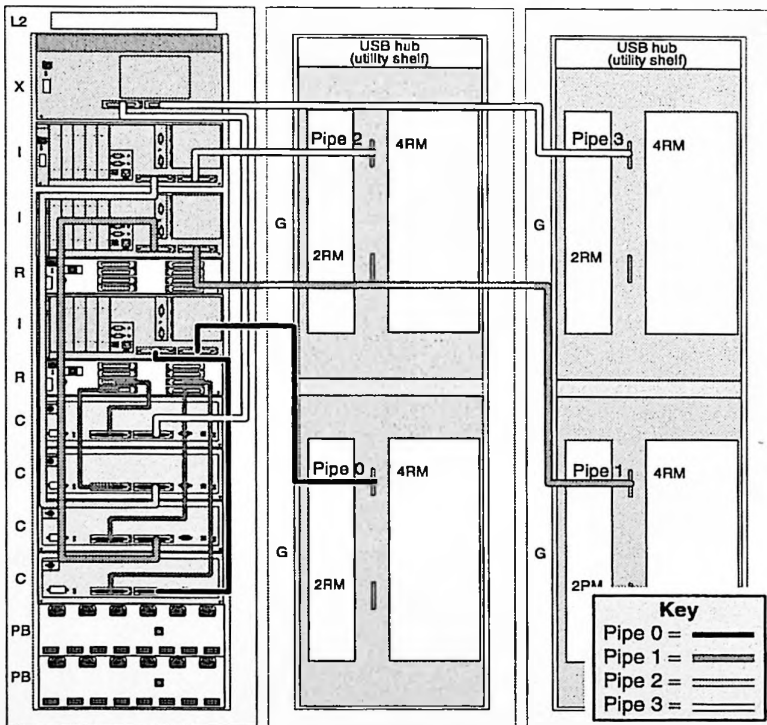


Figure 2-5 Example Pipe Sequencing

- In partitioned systems, pipe sequence restarts at 0 in each partition.

2.7 Suggested Pipe Configuration

Graphics pipes are assigned to the system based on the NASIDs of the C bricks to which they connect.

Configure pipes so that:

- Pipe 0 is the 4RM pipe in the lower section of the lowest-numbered graphics rack.
- The lower-numbered pipe in a G brick is the 4RM pipe (upper port), and the higher-numbered pipe in a G brick is the 2RM pipe (lower port).
- For systems where all graphics racks are adjacent (as shown in Figure 2-6), pipe numbers increase sequentially as follows:
 1. From the lower section of the lowest-numbered graphics rack,
 2. To the lower section of the highest-numbered graphics rack,
 3. To the upper section of the lowest-numbered graphics rack,
 4. To the upper section of the highest-numbered graphics rack.

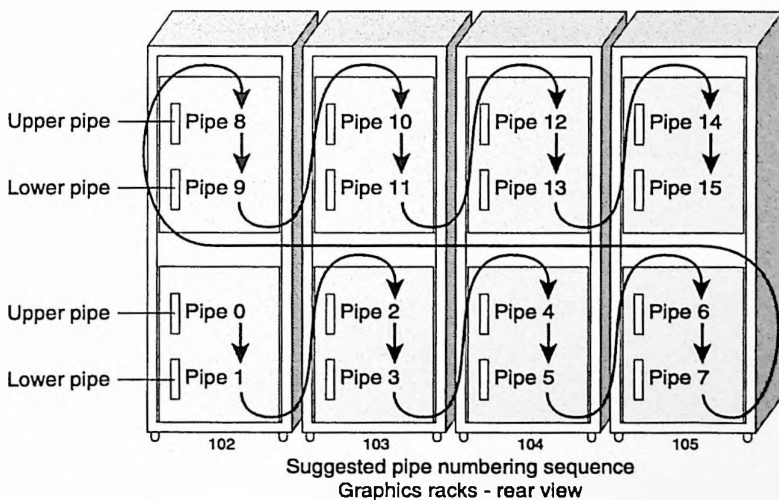


Figure 2-6 Suggested Pipe Sequence with All Pipes in Adjacent Racks

- For systems where graphics racks are not adjacent (as shown in Figure 2-7), pipe numbers increase sequentially as follows:
 - From the lower section of the lowest-numbered graphics rack,
 - To the lower section of the highest-numbered graphics rack in that quadrant,
 - To the upper section of the lowest-numbered graphics rack,
 - To the upper section of the highest-numbered graphics rack in that quadrant,
 - To the lower section of the lowest-numbered graphics rack in the next quadrant,
 - To the lower section of the highest-numbered graphics rack,
 - To the upper section of the lowest-numbered graphics rack in that quadrant,
 - To the upper section of the highest-numbered graphics rack.

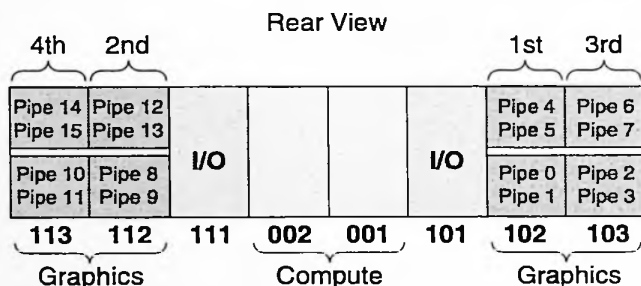


Figure 2-7 Suggested Pipe Sequence with Graphics Racks on Both Ends

2.8 Checking Actual Pipe Numbering

To determine which C bricks control the graphics pipes, look at the hwgraph files in */hw/gfx*.

Example

In the 4-pipe system in Figure 2-5, the */hw/gfx* output would like this:

```
firehouse 8# ls -l /hw/gfx
total 0
lrw----- 1 root sys 34 Feb 15
13:23 0 -> /hw/module/001c07/Ibrick/xtalk/10
lrw----- 1 root sys 34 Feb 15
13:23 1 -> /hw/module/001c10/Ibrick/xtalk/10
lrw----- 1 root sys 34 Feb 15
13:23 2 -> /hw/module/001c13/Ibrick/xtalk/10
lrw----- 1 root sys 34 Feb 15
13:23 3-> /hw/module/001c17/Xbrick/xtalk/10
      |           |           |
      Pipe number  C brick location  Xtown2 port on I/O brick
                                   (XI010; widget 10)
```

2.9 Pipe and X Server Numbering

2.9.1 hostname:0.0 Described

The components of the expression *hostname:0.0* signify the following:

- *hostname* = The name of the host machine.
- *:0* = The X server number (each graphics seat [keyboard, mouse, and monitor] needs an associated X server).
- *.0* = The Xscreen controlled by that X server. Xscreen .0 represents the first pipe assigned to an X server.

Table 2-2 Example Display Variables for 2 Graphics Seats, 2 Pipes Each

Display Variable	Physical Pipe Number
firehouse:0.0	0
firehouse:0.1	1
firehouse:1.0	2
firehouse:1.1	3

2.9.2 The Xservers File: Configuring Graphics Seats and Assigning Keyboard and Mouse Control

1. Edit the `/usr/lib/x11/xdm/Xservers` file to assign physical pipe numbers to each X server.
2. Restart the X server(s).

Example 1

For a system with 1 graphics pipe, the *Xservers* file contains data similar to the following:

```
:0 secure /usr/bin/X11/X -bs -nobitscale -c -pseudomap  
4sight -solidroot sgilightblue -cursorFG red -cursorBG white
```

Example 2

For a system with 3 graphics seats—where seat 0 has 3 graphics pipes, seat 1 has 2 pipes, and seat 2 has 1 pipe—the *Xservers* file contains data similar to the following:

```
:0 secure /usr/bin/X11/X : 0 -boards 0,1,2 -devdir /dev/input  
:1 secure /usr/bin/X11/X : 1 -boards 3,4 -devdir /dev/input1  
:2 secure /usr/bin/X11/X : 2 -boards 5 -devdir /dev/input2
```

X server (seat) boards = physical pipe Keyboard/mouse set

Physical pipe numbers

Table 2-3 lists the display variables and physical pipe numbers associated with the *Xservers* file shown above in Example 2.

Table 2-3 Display Variables and Pipe Numbers Associated with Example 2

Display Variable	Physical Pipe Number	Controlling Keyboard and Mouse Set
powerhouse:0.0	0	/dev/input
powerhouse:0.1	1	/dev/input
powerhouse:0.2	2	/dev/input
powerhouse:1.0	3	/dev/input1
powerhouse:1.1	4	/dev/input1
powerhouse:2.0	5	/dev/input2

2.9.3 Pipe Failure and Pipe Numbering

At boot, pipe numbers are assigned sequentially and never skipped.

If a pipe fails to come up at boot, that pipe's number is assigned to the next functional pipe in the sequence. Because the `/etc/ioconfig.conf` file is unaffected, this shift causes a discrepancy between the pipe output and the input devices (keyboard/mouse).

Example

In a 3-pipe system, the 2nd pipe (normally Pipe 1) fails to function at boot. As a result, the system assigns pipe number 1 to the 3rd pipe.

As shown in Figure 2-8, the keyboard and mouse at the 2nd seat now control the 3rd pipe (now Pipe 1). The monitor for the failed 2nd pipe is blank. The mouse and keyboard at the 3rd seat are inactive.

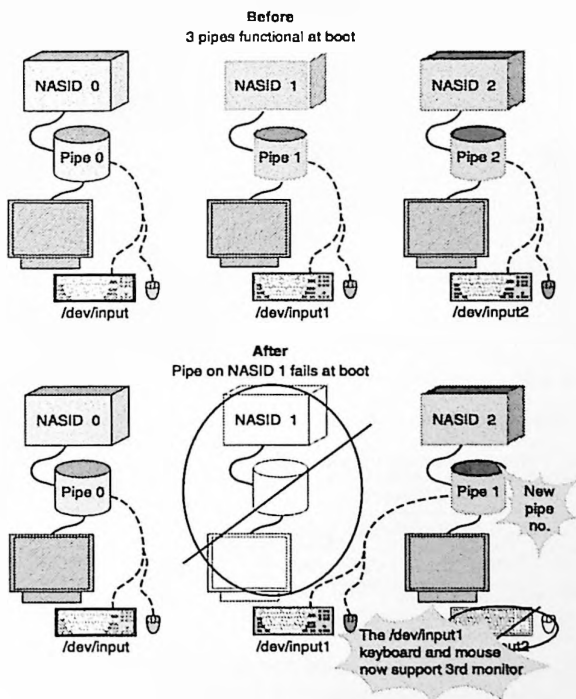


Figure 2-8 Numbering Before and After Pipe Failure at Boot – Example

2.10 Recommended I/O-bricks-to-G-brick Connections

Connect pipes in the lower section of a graphics rack to I/O bricks in the lower section of the I/O rack.

Connect pipes in the upper section of a graphics rack to I/O bricks in the upper section of the I/O rack.

For systems with 2 I/O racks per section of adjacent graphics racks (such as in Figure 2-9):

- Connect the pipes in the first two graphics racks (for example, 103 and 104) to bricks in the lower-numbered I/O rack (such as 101).
- Connect the pipes in the next two graphics racks (for example, 105 and 106) to bricks in the higher-numbered I/O rack (such as 102).

In general, connect I/O bricks to pipes in the following order:

1. I bricks
2. X bricks

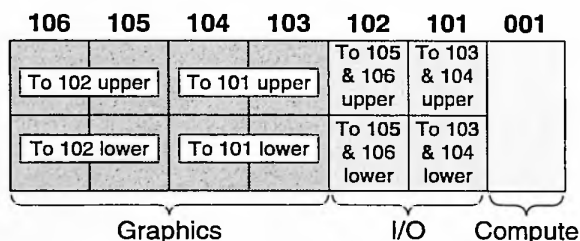


Figure 2-9 I/O Brick and Pipe Relation

Table 2-4 Suggested I/O-to-pipe Cabling: 1 I/O Rack in a Section

I/O Slot	Pipe	Suggested Pipe Location	I/O Slot	Pipe	Suggested Pipe Location
U16	0	4RM of 102 lower	U04	4	Varies
U04	1	Varies (2RM of 102 lower)	U08	5	Varies
U08	2	Varies	U12	6	Varies
U12	3	Varies	U32	7	Varies

Note: Applies to I and X bricks.

Table 2-5 Suggested I/O-to-pipe Cabling: 2 I/O Racks in a Section

I/O Rack	I/O Slot	Pipe
<i>Connect bricks in I/O Rack 101 to pipes in the lower G bricks of the first two graphics racks.</i>		
101	U16	0 (Lower 4RM, 1st graphics rack)
	U04	1 (Lower 2RM, 1st graphics rack)
	U08	2 (Lower 4RM, 2nd graphics rack)
	U12	3 (Lower 2RM, 2nd graphics rack)
<i>Connect I/O Rack 102 to pipes in the lower G bricks of the second two graphics racks, if required.</i>		
102	U16	Next assigned port
	U04	Next assigned port
	U08	Next assigned port
	U12	Next assigned port
<i>Connect I/O Rack 102 to pipes in the upper G bricks of the first two graphics racks.</i>		
102	U20	Next assigned port
	U24	Next assigned port
	U28	Next assigned port
	U32	Next assigned port
<i>Connect I/O Rack 102 to pipes in the upper G bricks of the second two graphics racks, if required.</i>		
102	U20	Next assigned port
	U24	Next assigned port
	U28	Next assigned port
	U32	Next assigned port
<i>Repeat these four patterns for the next section.</i>		

Note: Applies to I and X bricks.

Note: Xtown2 ports on new I and X bricks are set to 400 MHz.

2.11 Upgrade Notes

Racks and G bricks for upgrades ship with 2-meter (green) Xtown2 cables, which may be too short for some configurations. When the customer orders new graphics racks or G bricks, the customer may need to order additional, longer Xtown2 cables to replace the 2-meter cables.

If you must change the old pipe sequence during an upgrade, consider recabling the old C-brick-to-I/O-brick connections, rather than recabling Xtown2 cables between I/O bricks and pipes.

2.12 Power Cabling

The G brick has its own power supply, which connects to facility power—not to the power bay.

The power supply is located on the front side of the midplane.

The utility-tray devices (such as Ethernet hub, USB hub, and router) connect to the power strip in the graphics rack.

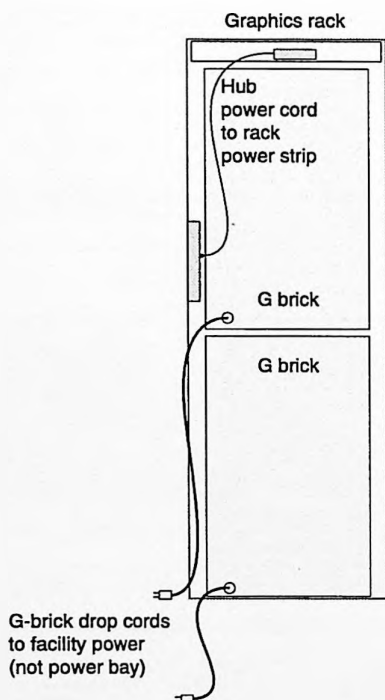


Figure 2-10 Power Cabling

2.13 Keyboard and Mouse Configuration

Refer to "Keyboard, Mouse, and Hubs" on page 1-23.

Commands, Tools, and Troubleshooting

Chapter Contents

3.1	Starting and Stopping Graphics	3-2
3.2	Graphics Maintenance Tools	3-2
3.2.1	Diagnostic Testing (<i>irsaudit</i>)	3-2
3.2.2	Video Format Combiner (<i>ircombine</i>)	3-3
3.2.3	Maintaining the EEPROM (<i>ireeprom</i>)	3-4
3.2.3.1	Restoring or Upgrading the EEPROM Code	3-4
3.2.3.2	Restoring a Default Combination	3-5
3.2.4	Changing the Video Format with <i>setmon</i>	3-5
3.3	Displaying the Graphics Subsystem Information	3-6
3.4	Displaying the Graphics Topology	3-6
3.5	Running Boundary Scan on Graphics Hardware	3-6
3.6	L1, L2, PROM Monitor, and POD commands	3-6
3.7	Troubleshooting Graphics	3-7

3.1 Starting and Stopping Graphics

Rebooting the system starts the graphics system, but you can manually start graphics by using the commands listed in Table 3-1.

Table 3-1 Starting and Stopping Graphics

Command	Function
<code>/usr/gfx/stopgfx</code>	Stops all system graphics from a dumb or remote terminal
<code>/usr/gfx/startgfx</code>	Starts all system graphics from a dumb or remote terminal
Shift-Ctrl-F12-/ (Vulcan Death Grip)	Stops and restarts the X server that is associated with this keyboard
Use / on numeric keypad	

3.2 Graphics Maintenance Tools

3.2.1 Diagnostic Testing (*irsaudit*)

- You cannot run *irsaudit* from the console (pipe and seat) that you are testing; *irsaudit* takes direct control of the graphics hardware.
- Use `/usr/gfx/stopgfx` to stop the pipe's X server before you run *irsaudit*.
- To target a specific pipe, include `PIPE=n` on the command line (where *n* is the pipe number). The default is pipe 0.
- To test multiple pipes individually, run separate instances of *irsaudit* and direct the output to different log files. For example:
`irsaudit PIPE=0 LOG=0.log&`
`irsaudit PIPE=1 LOG=1.log&`
- By default, *irsaudit* tests all boards and options in the pipe (GE, RM, DG). The following command-line options limit the testing:
 - `-ge` - runs the GE board tests
 - `-rm` - runs the RM board tests
 - `-dg` - runs the DG board tests
 - `-dplex` - runs the DPLEX board tests (for DG5-DPLEX)
 - `-gvo` - runs the GVO board tests (for DG5-GVO)
 - `-tvo` - runs the TVO board tests (for DG5-TVO)
- Use `/usr/gfx/startgfx` to restart the pipe's X server after running *irsaudit*.

- *irsaudit* was updated to support InfiniteReality3. There are no new tests.
- *irsaudit* creates two files in `/var/adm/crash/diags/gfx/IR` that *availmon* uses to track the system:

`irsaudit_number.rslt` and `irsaudit_number.sum`

3.2.2 Video Format Combiner (*ircombine*)

Each instance of *ircombine* configures one pipe. To configure multiple pipes, run *ircombine* for each pipe.

To invoke *ircombine* in—and target changes to—your current display, enter:

```
/usr/gfx/ircombine
```

To invoke *ircombine* in your current display but target changes to a different pipe on the system, enter:

```
/usr/gfx/ircombine -target :x.s
```

(where *x* is the X display and *s* is the X screen of the target)

To invoke *ircombine* on a remote Onyx 3000 series (or Onyx2) system but display the GUI on your local X Window System:

- Connect to the remote system (UUCP or TCP/IP connection)
- Log in as root
- Enter:

```
/usr/gfx/ircombine -target Onyx:x.s -display  
localsystem:0.0
```

where *onyx:* is the name of the Onyx 3000 series system; *x* is the X display; *s* is the target X screen; and *localsystem:0.0* is the remote display

3.2.3 Maintaining the EEPROM (*ireeprom*)

The `/usr/gfx/KONA/bin/ireeprom` command enables you to reflash the GE EEPROM microcode. Table 3-2 lists the *ireeprom* command options:

Table 3-2 *ireeprom* Options

Option	Function
-w	Write EEPROM from file
-r	Read EEPROM into file
-e	Erase EEPROM NOTE: Do not use this option!
-c	Erase EEPROM video combination
-v	Verify contents of EEPROM
-i	Print EEPROM information
-F	Forcibly load older revision
-h	Print this help message
-p	<pipenum>: Specify pipe number
-f	<datafile>: Specify microcode datafile

3.2.3.1 Restoring or Upgrading the EEPROM Code

Use the following procedure to restore or upgrade the EEPROM code:

1. From an alternate console, log in to the system as root.
2. Change to the `/usr/gfx/KONA/bin` directory:

```
cd /usr/gfx/KONA/bin
```
3. Before you flash the EEPROM, use the `./ireeprom -i` command to check the microcode. If the microcode is good, you should see output similar to the following:

```
# ./ireeprom -i
==== Pipe 0 ====
Manufacturer code:      0x1f
Device code:           0xc4
State:                 Good
Revision:              114|
timestamp:             Fri Dec 30 15:46:24 2000
Code+data length:      0x1caac
Checksum:              0x009698fd
```

For multipipe systems, use the `-p` option to specify the correct pipe.

4. Overwrite the microcode by using the following command. If you use an older or identical version of the microcode to overwrite the EEPROM microcode, you must use the `-F` option.

```
./ireeprom -w -F -f /usr/gfx/ucode/KONA/tport.bin
```

5. Restart the graphics subsystem to download the new microcode. For example, from a remote terminal:

```
(/usr/gfx/stopgfx; sleep 2; /usr/gfx/startgfx)&
```

3.2.3.2 Restoring a Default Combination

If *setmon* hangs when you try to restore a default combination, you can use `ireeprom -c` to erase bad combination files in the EEPROM and restore a default combination. If you download a combination file that causes the monitor to lose sync, the screen goes black and you cannot log in to the graphics terminal.

1. Use an alternate console to log in to the system as root.
2. Change to the `/usr/gfx/KONA/bin` directory:

```
cd /usr/gfx/KONA/bin
```

3. Erase the EEPROM video combination:

```
./ireeprom -c
```

4. Restart the graphics subsystem to download the default video combination. For example, from a remote terminal:

```
(/usr/gfx/stopgfx; sleep 2; /usr/gfx/startgfx)&
```

The system will restart with a default combination of 1280x1024_60.

3.2.4 Changing the Video Format with *setmon*

To change or restore the default video format, enter the `/usr/gfx/setmon` command as root. Then stop and restart graphics.

Example:

```
# /usr/gfx/setmon -x 1280x1024_60
# /usr/gfx/stopgfx
# /usr/gfx/startgfx
# /usr/gfx/gfxinfo
```

To work, *setmon* must access the X server. If the X server is locked, *setmon* will hang.

3.3 Displaying the Graphics Subsystem Information

Use the following command to display graphics subsystem information:

```
# /usr/gfx/gfxinfo
```

Use the `-v` option for verbose output.

Use the `-vv` option for very verbose output.

3.4 Displaying the Graphics Topology

Use the following command to display the physical layout of the pipes and input devices:

```
# /usr/gfx/gfxtopology
```

Note: The `/usr/gfx/gfxtopology` script may be included with IRIX 6.5.12 or a later release.

3.5 Running Boundary Scan on Graphics Hardware

For information on scan tests and the graphics hardware that they test, refer to *SGI 3000 Family Scan Tools*, 108-0263-00x.

3.6 L1, L2, PROM Monitor, and POD commands

For L1, L2, PROM Monitor, and POD commands, refer to Chapter 4 of the *Hardware Components Quick Reference* booklet, 108-0242-00x.

3.7 Troubleshooting Graphics

The tips in Table 3-3 and the following list of questions can help you isolate problems in the graphics subsystem.:

Table 3-3 Symptom and Cause Matrix

Symptoms	Possible Cause	Diagnosis and Correction Tips
Host system cannot communicate with the graphics subsystem or behaves as if the graphics hardware does not exist	- C brick - I/O brick - Xtown2 cable - Ktown2 board - G-brick midplane - GE board	- Check connections - Run <i>irsaudit</i> * - Check console messages and logs - Reseat boards - Swap cables - Swap boards
PROM monitor takes an exception or hangs during the power-on boot process in the initializing-software-and-devices stage	- GE board - DG5 board	- Pull keyboard cable and try to reboot system (if the server side comes up, it is a graphics problem)
IRIX takes an exception or hangs when starting graphics	- GE board - DG5 board	- Reseat boards - Replace failed board
Garbled text on display	- GE board - GE EEPROM microcode	- Run <i>irsaudit</i> * - Use <i>ireeprom -i</i> to verify microcode checksum (reload microcode if needed) - Reseat boards - Replace failed board
Window cannot open	- GE board - GE EEPROM microcode - Bad Xwindows software or configuration file	- Run <i>irsaudit</i> * - Use <i>ireeprom -i</i> to verify microcode checksum (reload microcode if needed) - Restore faulty software image - Isolate problem to system or user - Reseat boards

Table 3-3 Symptom and Cause Matrix (continued)

Symptoms	Possible Cause	Diagnosis and Correction Tips
Pixel drop out, such as - Vertical lines (solid or dashed) - Evenly spaced holes	- RM board (faulty board, bent pins, or bad flex connector) – raster manager or texture memory problem	- Run <i>irsaudit</i> * - Remove all RM boards except board 0 to see if the problem goes away - Reseat boards - Swap boards - Replace failed board Note: The pipe cannot function without the board in RM slot 0.
Blank or unsynchronized monitor display	- DG5 - Monitor cable - Format problem in GE EEPROM or DG5 VOC registers - Corrupt format - Format exceeds the monitor's CRT resolution or scan frequency specifications (invalid VOC parameters)	- Run <i>irsaudit</i> * - Reseat boards - Replace faulty board or cable - Use monitor self-test procedures to isolate - Load valid format with <i>ireeprom -c</i> command - Power cycle the monitor
Wrong colors or missing color	- GE board - DG5 board - Monitor cable - Monitor color settings	- Run <i>irsaudit</i> * - Check monitor color settings - Replace faulty board, cable, or monitor - Secure loose cable

* *irsaudit* cannot diagnose the Xbridge-to-GE path that the host system uses to communicate with the graphics subsystem. If this path is failing, look for a console warning message.

Questions to Ask

- Has someone modified any files?
- Has someone installed new software or upgrades?
- Has someone installed new hardware?
- Does *SYSLOG* show any graphics errors?
- Do you have a similar problem on other systems?
- Have you reseated the graphics boards and cables?
- Is the display color or sync bad?

If so, have you:

- Checked the cable connection from the computer to the monitor?
 - Checked the monitor's Input Select switch?
 - Tried a different monitor or cable?
 - Used *setmon* or *ireeprom -c* to reset the monitor resolution and sync settings to defaults?
 - Used *ireeprom -i* to verify the graphics microcode checksum?
- If the X session hangs or core dumps during startup, have you tried initiating an X session manually from the command line?

To do so:

1. Enter the following command:

```
xsgi -ac&
```

2. Set the DISPLAY variable to :0.0.

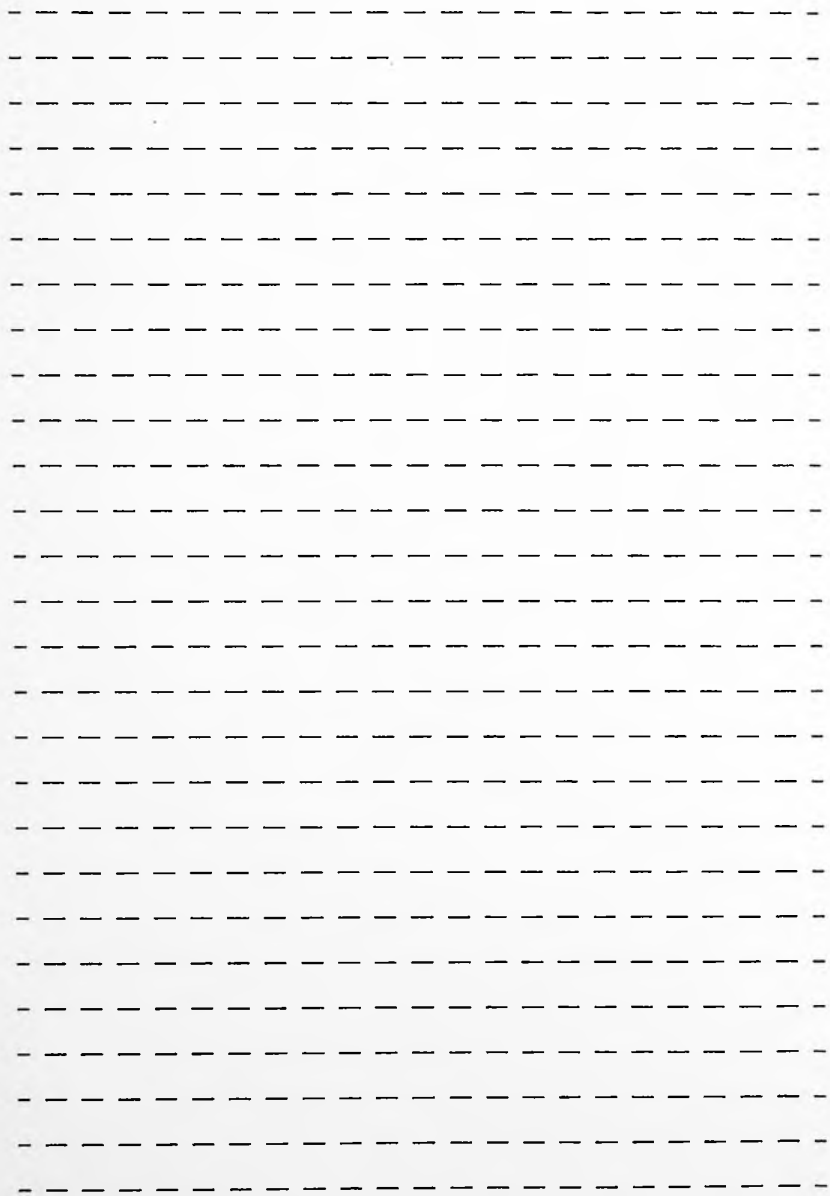
3. Enter the following command:

```
4Dwm&
```


Notes

This image shows a full page of a document template. It consists of a series of evenly spaced, horizontal black lines on a white background. The lines are uniform in length and thickness, extending across the entire width of the page. There are no margins, text, or other markings present.

Handwriting practice lines consisting of multiple rows of dashed lines for tracing and writing practice.



Index

Numbers

13W3

- monitor cables, 1-20
- pinouts on DG5, 1-16

A

Audio

- breakout cable, 1-28
- PCI card, 1-27 – 1-29
- pinouts, 1-29
- speakers, 1-29 – 1-30

B

- Bandwidth, calculating, 1-17
- Bay numbering, 2-4
- Boundary scan, 3-6
- Breakout cable
 - audio, 1-28
 - XTHD, 1-32

C

Cable

- audio breakout, 1-28
- extender hub, 1-25
- part numbers, 1-35 – 1-36
- power, 2-14
- standard USB hub, 1-23
- XTHD breakout, 1-32

Cables

- L2, 1-34
- monitor, 1-20

Combination

- restoring, 3-5

Commands

- boundary scan, 3-6
- DIVO, 1-31
- gfxinfo, 3-6
- gfixtopology, 3-6
- ireeprom, 3-4 – 3-5
- irsaudit, 3-2
- setmon, 3-5
- startgfx, 3-2
- start X session, 3-9
- stopgfx, 3-2
- Vulcan death grip, 3-2

Configuration

- checking pipe numbers, 2-9
- DIVO, 1-31
- examples, 2-3 – 2-5
- input devices, 1-27
- overview, 2-2
- pipe, 2-7 – 2-8
 - and I/O brick, 2-12
 - and pipe failure, 2-11
 - and X server, 2-9
 - with ircombine, 3-3
- rack numbering, 2-4
- rack placement, 2-5
- rules, 2-2
- seat, 2-10

Controller numbers/IDs, 1-26

Controllers

- L1
 - basic functions, 1-34
 - components, 1-33

D

DDO2

- image, 1-15

- notes, 1-12

Death grip, Vulcan, 3-2

Device IDs, 1-26

/dev/input, 1-26

DG5 boards

- board options, 1-10

- calculating bandwidth, 1-17

- connectors, standard, 1-11

DDO2

- image, 1-15

- notes, 1-12

DG5-2, 1-10

- image, 1-13

DG5-8

- image, 1-14

- notes, 1-12

DPLEX

- connectors, 1-11

- image, 1-14

- notes, 1-12

GVO

- connectors, 1-11

- image, 1-15

monitor connector pinouts, 1-16

TVO

- connectors, 1-11

- image, 1-16

- notes, 1-13

Diagnostics

- DIVO, 1-31

- irsaudit, 3-2

- XT-HD, 1-32

Digital media, XIO options,

- 1-31 – 1-32

Displaying graphics information,

- 3-6

Displaying graphics topology, 3-6

Display. See Monitor

DIVO, 1-31

DIVO-DVC, 1-31, 1-32

DPLEX

- connectors, 1-11

- image, 1-14

- notes, 1-12

E

EEPROM, updating, 3-4 – 3-5

F

Flat panel. See Monitor

Format, updating, 3-5

G

G brick

- board locations, 1-9

- compatible boards, 1-9

- components, 1-8

- dimensions, 1-7

- L1 to L2 connection, 1-23

- power cabling, 2-14

- vs. KCAR (Onyx2), 1-7, 1-8

GE boards

- image, 1-19

- InfiniteReality configurations, 1-9

Graphics

- board sets, 1-9

- starting and stopping, 3-2

- troubleshooting, 3-7 – 3-8

GVO

- connectors, 1-11

- image, 1-15

H

hostname:0.0, 2-9

Hub

- LEX/REX extender, 1-24 – 1-25
- power cabling, 2-14
- standard, 1-23

/hw/gfx, 2-9

hwgraph, 1-23

I

IDs, device, 1-26

InfiniteReality

- board types, 1-9
- configurations, 1-9

Input devices

- and X server, 1-26
- configuring, 1-27
- IDs, 1-26

I/O brick

- pipe connections, 2-12
- USB ports, 1-23

ioconfig.conf, 1-26

- and pipe failure, 2-11

ircombine, 3-3

ireeprom command, 3-4 – 3-5

irsaudit, 3-2

IR. See InfiniteReality

K

Keyboard

- and pipe failure, 2-11
- assigning seats, 2-10
- configuring, 1-27
- ID, 1-26

Ktown2, 1-17

L

L1

- basic functions, 1-34
- components, 1-33

LEDs

- speaker, 1-30

M

Monitor

- cabling options, 1-20 – 1-21
- connecting a flat panel, 1-22
- connector pinouts on DG5, 1-16
- options for Onyx 3000, 1-20

Mouse

- and pipe failure, 2-11
- assigning seats, 2-10
- configuring, 1-27
- ID, 1-26

N

NASIDs

- and controller numbers, 1-26
- and pipe numbering, 2-6

Numbering

- input devices, 1-26
- pipe, 2-6
 - and I/O brick, 2-12
 - and pipe failure, 2-11
 - and X server, 2-9
 - checking, 2-9
 - suggested, 2-7 – 2-8
- rack, 2-4

Numbers, part, 1-35 – 1-36

O

Onyx3, 1-2

See also G brick
Onyx 3000
block diagram, 1-3 – 1-4
components, 1-2
configuration
examples, 2-3 – 2-5
overview, 2-2
rack numbering, 2-4
rack placement, 2-5
rules, 2-2
monitor options, 1-20
rack
front, 1-5
rear, 1-6

P

Part numbers, 1-35 – 1-36
PCI card
audio, 1-27 – 1-29
SIO, 1-31
Pinouts
DG5 monitor connector, 1-16
video sync and digital audio, 1-29
Pipe
I/O brick connections, 2-12
numbers
and pipe failure, 2-11
and X server, 2-9
assigning, 2-6
checking, 2-9
suggested, 2-7 – 2-8
Power cabling, 2-14
Problems, troubleshooting, 3-7 – 3-8
Procedures
calculating bandwidth, 1-17
configuring DIVO, 1-31
configuring input devices, 1-27
configuring seats, 2-10
connecting flat panel, 1-22

displaying graphics information,
3-6
displaying graphics topology, 3-6
running boundary scan, 3-6
starting and stopping graphics,
3-2
testing DIVO, 1-31
testing with irsaudit, 3-2
testing XT-HD, 1-32
updating EEPROM, 3-4 – 3-5
updating video format, 3-5
using ircombine, 3-3

R

Rack
numbering, 2-4
placement, 2-5
RM boards
image, 1-18
InfiniteReality configurations, 1-9
slot locations, 1-9

S

scantest, 3-6
setmon command, 3-5
Shift-Ctrl-F12-/ , 3-2
SIO PCI card, 1-31
slit, 3-6
Slot numbering, 2-4
Solving problems, 3-7 – 3-8
Sound card, PCI, 1-27 – 1-29
Speakers, 1-29 – 1-30
Starting graphics, 3-2
Stopping graphics, 3-2
Sync
pinouts on audio card, 1-29
XT-HD, 1-32

T

Tests

- DIVO, 1-31
- irsaudit, 3-2
- XT-HD, 1-32

Tools

- ircombine, 3-3
- irsaudit, 3-2

Troubleshooting, 3-7 – 3-8

TVO

- connectors, 1-11
- image, 1-16
- notes, 1-13

U

USB ports

- I brick, 1-23
- LEX/REX hub, 1-24
- standard hub, 1-23
- /usr/diags/DIVO/bin, 1-31
- /usr/diags/XTHD/bin/xthd_confidence, 1-32
- /usr/dmedia/bin/DIVO, 1-31
- /usr/gfx/gfxinfo, 3-6
- /usr/gfx/gfxtopology, 3-6
- /usr/gfx/KONA/bin/ireeprom, 3-4 – 3-5
- /usr/gfx/setmon, 3-5
- /usr/gfx/startgfx, 3-2
- /usr/gfx/stopgfx, 3-2
- /usr/sbin/vcp, 1-31

V

VCP, 1-31

Video format, updating, 3-5

Video format combiner (ircombine), 3-3

Video library control panel, 1-31

Vulcan death grip, 3-2

X

Xscreen, and host name, 2-9

X server

and pipe numbering, 2-9

input device numbers, 1-26

See also Xservers file

Xservers file

configuring seats, 2-10

See also X server

X session, starting from command line, 3-9

XT-HD, 1-31, 1-32

Xtown2, 1-7



